

PART III MAGNETOSTATICS

LECTURE VI. THE MAGNETIC FIELD

VI.1 LECTURE OUTLINE

Electromagnetism is the combination of electric and magnetic phenomena. In this lecture, we will define the magnetic field as it acts on moving electric charges and by extension on conductors carrying DC currents. The direct application of such an interaction is the electric motor principle.

Specific objectives

At the end of this lecture, you should be able to:

1. Define the magnetic field vector and its main properties
2. Describe the lines of the magnetic and distinguish them from the lines of electrostatic fields
3. Compute the force acting on a charge in a magnetic field
4. Describe the force acting on a conductor carrying current
5. Explain and apply the electric motor principle

VI.2. THE MAGNETIC FIELD

VI.2.1 Introduction

A charged plastic rod produces a vector field- the electric field E at all points in the space around it. Similarly a magnet produces a vector field B at all points in the space around it. An electric charge sets up an electric field that can affect other electric charges. We might expect that a magnetic charge sets up a magnetic field that can affect other magnetic charges. Although such magnetic charges, called **magnetic monopoles** are predicted by certain theories, their existence has not been confirmed.

Magnetic fields are set up in two ways:

1. Moving electrically charged particles, such as a current in a wire
2. Elementary particles such as electrons have intrinsic magnetic field around them

VI.1.2 Definition of B

The **electric field** $\vec{E}$ at a point is tested by placing a test particle of charge q_0 at rest at that point and measuring the

electric force F_E acting on the particle and we define $\vec{E}$ as:
$$\vec{E} = \frac{\vec{F}_E}{q_0}$$

To define the **magnetic vector** $\vec{B}$, we fire a **moving charged particle** with different velocities in various directions. After many trials we will notice that three properties:

1. There is a particular direction for which the force acting on the particle is zero. We call it “**zero-force axis**”
2. For other directions of $\mathbf{v}$, the magnitude of F_B is always proportional to $v \sin \phi$, where ϕ is the angle between the zero-force axis and the direction of $\mathbf{v}$.
3. The direction of F_B is always perpendicular to the direction of $\mathbf{v}$.

To reflect these proprieties we then define a magnetic field $\vec{B}$ to be a vector quantity whose:

- (i) Direction is along the zero-force axis
- (ii) The magnitude of the force is maximum when $\mathbf{v}$ is perpendicular to the zero-force axis, such that:

$$B = \frac{F_B}{|q_0|v},$$

where q_0 is the test charge

We can summarize all these results with the following vector equation:

$$\vec{F}_B = q[\vec{v} \times \vec{B}] \quad (\text{VI.1})$$

The magnitude of the force acting on a charged particle in a magnetic field is:

$$F_B = |q|vB \sin \phi \quad (\text{VI.2})$$

where ϕ is the angle between the velocity of particle and the field.

As a vector product the force F_B acting on a charged particle moving with velocity $\mathbf{v}$ through in a magnetic field $\mathbf{B}$ is always perpendicular to both $\mathbf{v}$ and $\mathbf{B}$

The **SI unit** for B is **Tesla, T**:

$$[B] = \frac{1\text{Newton}}{1\text{Coulomb} \cdot \frac{1\text{metre}}{\text{second}}} = 1 \frac{N}{A \cdot m} \quad (\text{VI.3})$$

The **Gauss, G** is also used: $1G = 10^{-4}T$

The unit of magnetic field vector was named in the honour of Engineer Nikola Tesla* and the Gauss in the honour of Johan C.F. Gauss**

***Nikola Tesla (10 July 1856 – 7 January 1943)** was a *Serbian-American inventor, electrical engineer, mechanical engineer, physicist, and futurist* best known for his contributions to the design of the modern *alternating current (AC) electricity supply system*.

****Johann Carl Friedrich Gauss (30 April 1777 – 23 February 1855)** was a *German mathematician and physical scientist* who contributed significantly to many fields, including *number theory, algebra, statistics, analysis, differential geometry, geodesy, geophysics, electrostatics, astronomy and optics*

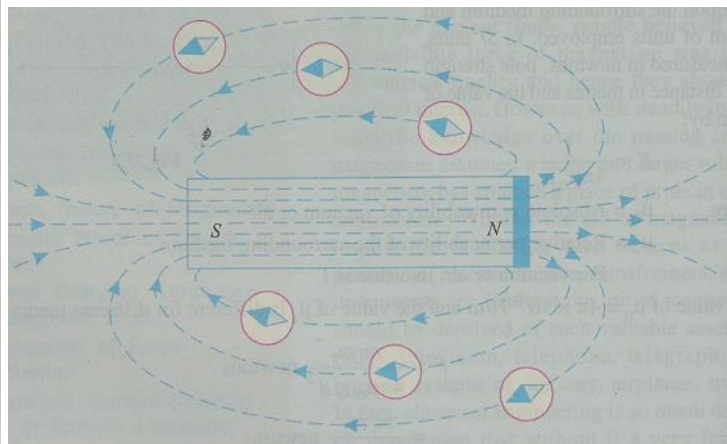
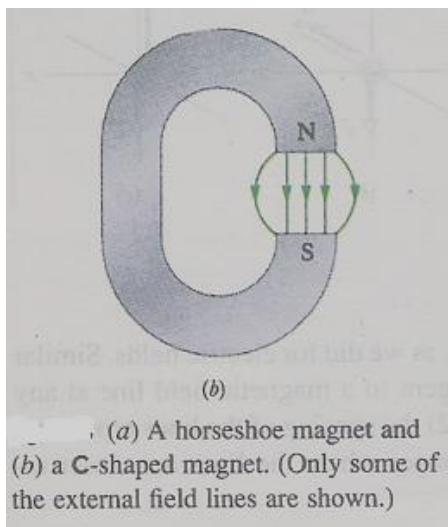
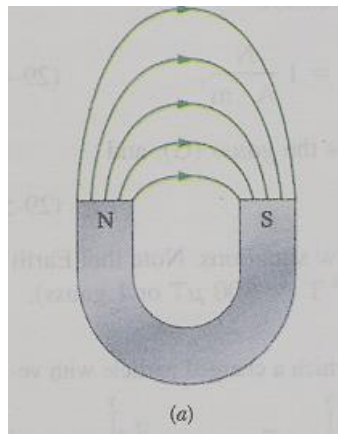
VI.3. MAGNETIC FIELD LINES

The magnetic field lines are plotted similarly to electric field lines:

1. The direction of the tangent to a magnetic field line at any point gives the direction of B at the point
2. The spacing of the lines represents the magnitude of B

Properties of magnetic field lines

1. Each magnetic field line forms a **closed loop**: outside the magnet the direction is from the north pole to south pole and it continues through the body of the magnet to form a closed loop
2. No two magnetic lines can intersect otherwise the field B would have two different directions at same point, what is not possible
3. Magnetic field lines are always ready to pass through iron or other magnetic materials in preference to passing through air even though their closed paths are made longer thereby



(c) Field of a bar magnet

Fig.VI.1. Configurations of magnetic field created by different types of magnets

VI.4. MAGNETIC FORCE ON A CURRENT-CARRYING WIRE

As we have seen, a magnetic field exerts a sideways force on a moving charge. Thus free electrons moving inside a current-carrying wire experience a force when the wire is placed in a magnetic field. This force must be transmitted

to the wire itself because the conduction electrons cannot escape sideways out of the wire. Fig.VI.2 (a) shows a vertical wire carrying no current and fixed in place at both ends extends through the gap between the vertical pole faces of a magnet. In Fig.VI.2 (b), a current is sent upward through the wire; the wire deflects on the right when the current is reversed in VI.2 (c) the wire deflects to the left

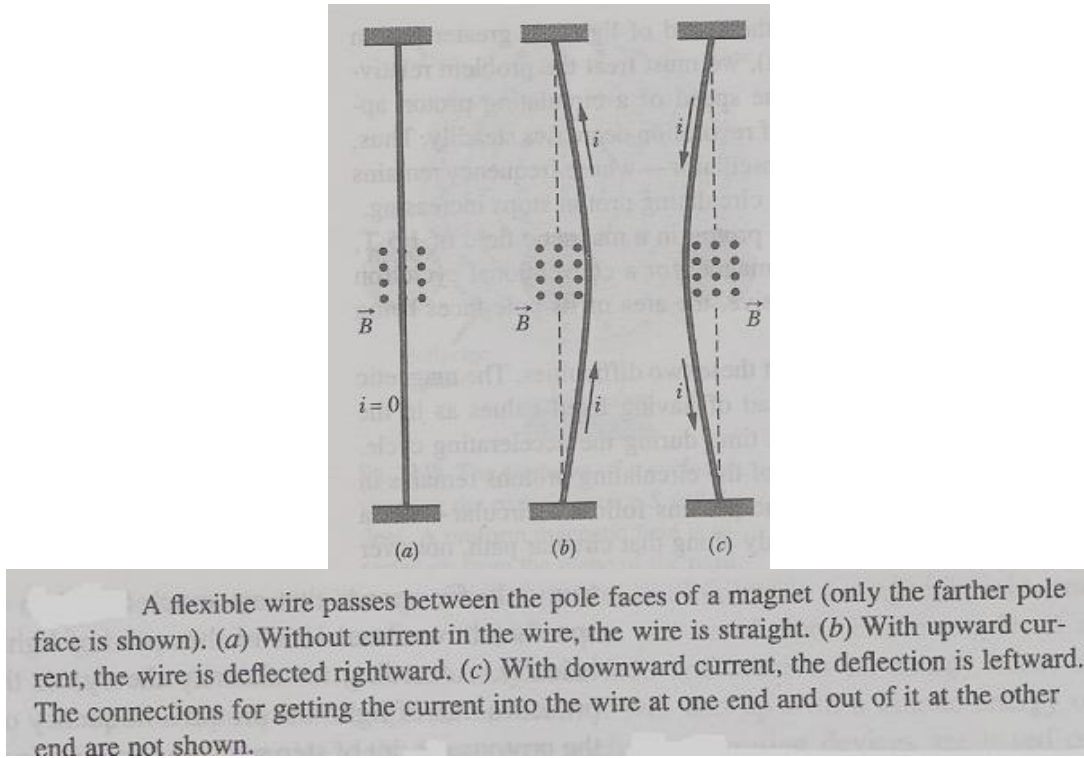


Fig.VI.2. A wire with current in a magnetic field

In Fig.VI.3, we consider a length L of a wire with current. All conduction electrons will drift past plane xx in a

time $t = \frac{L}{v_d}$ This corresponds to charge q :

$$q = it = i \frac{L}{v_d}$$

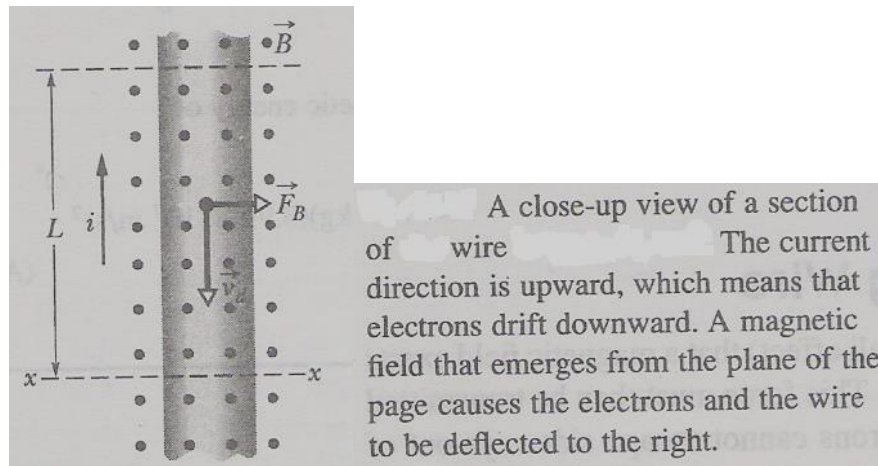
The total force acting on all these electrons:

$$F_B = qv_d B \sin \phi = \frac{iL}{v_d} v_d B \sin \frac{\pi}{2}$$

or $F_B = iLB$

Because L has a certain direction, this becomes:

$$\vec{F}_B = i[\vec{L} \times \vec{B}] \quad (VI.4)$$



VI.3. Force acting on an electron in a wire with current placed in a magnetic field

If the wire is not straight and the magnetic field not uniform, the force element $d\vec{F}_B$ acting on an element of length dL of the wire is:

$$d\vec{F}_B = i[d\vec{L} \times \vec{B}] \quad (\text{VI.5})$$

VI.5. TORQUE ON A CURRENT LOOP

Electric motors are based on the force that a magnetic field exerts on a wire carrying current. Fig.VI.4 shows a simple motor consisting of a single current carrying loop immersed in a magnetic field B

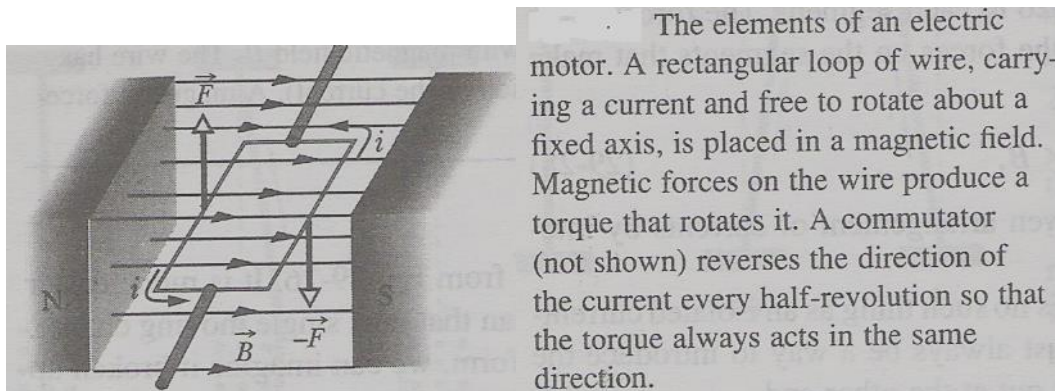
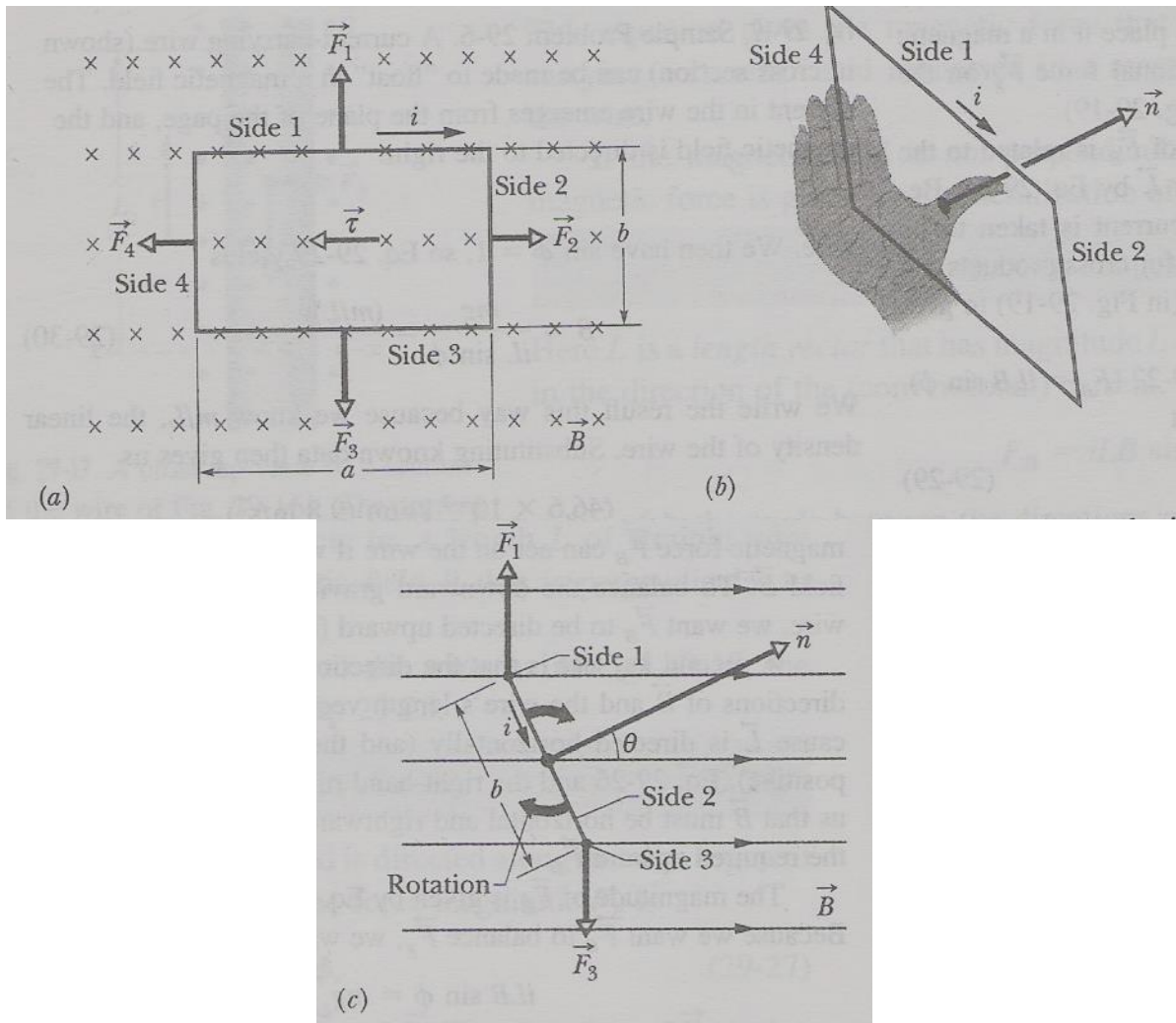


Fig.VI.4. Electric motor principle

The two magnetic forces F and $-F$ produce a torque on the loop tending to rotate it about its central axis. The production of the torque is explained in detail in Fig.VI.5.



A rectangular loop, of length a and width b and carrying a current i , is located in a uniform magnetic field. A torque $\vec{\tau}$ acts to align the normal vector $\vec{n}$ with the direction of the field. (a) The loop as seen by looking in the direction of the magnetic field. (b) A perspective of the loop showing how the right-hand rule gives the direction of $\vec{n}$, which is perpendicular to the plane of the loop. (c) A side view of the loop, from side 2. The loop rotates as indicated.

Fig.VII.5. Torque acting on a rectangular loop in a magnetic field

The net force on the loop is the vector sum of the forces acting on its four sides. For side 2 the vector L In equation (VI.4) points in the direction of the current and has magnitude b . The angle between L and B for side 2

at Fig.VII.5 (c) is $\left(\frac{\pi}{2} - \theta\right)$. Thus the magnitude of the force is:
$$F_2 = ibB \sin\left(\frac{\pi}{2} - \theta\right) = ibB \cos \theta$$

F_4 acting on side 4 has the same magnitude as F_2 but opposite direction and the two forces cancel out exactly.

For sides 1 and 3 things are different: L is perpendicular to B , so forces F_1 and F_3 have the same magnitude

$$F_1 = F_3 = iaB$$

These forces are opposite: they do not tend to move the loop up or down, but they do not share the same line of action so they produce a net torque. The torque tends to rotate the loop so as to align its normal vector $\vec{n}$ with the direction of the magnetic field B .

The torque τ' due to forces F_1 and F_3 :

$$\tau' = \left(iaB \frac{b}{2} \sin \theta \right) + \left(iaB \frac{b}{2} \sin \theta \right) = iaBb \sin \theta$$

Let us replace the single loop with a coil of N loops or turns tightly wound together. The total torque on the coil:

$$\tau = N\tau' = NiaBb \sin \theta = (NiA)B \sin \theta \quad (\text{VI.6}),$$

where $A = ab$, the area enclosed by the loop

In a motor, the current in the coil is reversed as $\vec{n}$ begins to line up with the field direction so that a torque continues to rotate. This automatic reversal of the current is done via a **commutator** that electrically connects the rotating coil with the stationary contacts on the wires that apply the current from source.

VI.6. MAGNETIC DIPOLE MOMENT

We can describe the current-carrying coil with a vector $\vec{\mu}$, the **magnetic dipole moment**. $\vec{\mu}$ has the direction of n :

$$\vec{\mu} = NiA\vec{n} \quad (\text{VI.7})$$

Unit SI of magnetic dipole moment: $[\mu] = \text{Ampere.m}^2$

The torque on a current-carrying loop in a magnetic field is:

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (\text{VI.8})$$

A loop with carrying current is called a **magnetic dipole** with a moment given by (VI.7)

Fig.VI.6 shows the orientation of the magnetic dipole moment produced by a loop with current

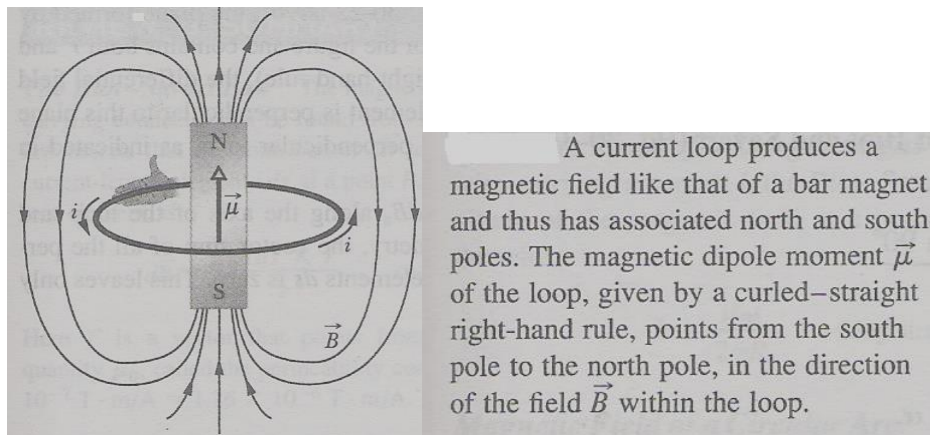


Fig.VI.6. A loop with current as a magnetic dipole

VI.7. MOTION OF CHARGED PARTICLES IN A MAGNETIC FIELD

As we have seen from the beginning, a moving charged particle experiences a force in a magnetic field. Magnetic fields are used to deflect and focus the electron beam in a TV tube and to separate elementary particles produced in high-energy particle accelerators.

Even though the Earth's magnetic field is relatively weak, it plays a part in protecting us from high-energy particles that bombard us from space

Fig.VI.7 shows two views of a positively charged particle with an initial velocity $\vec{v}$ perpendicular to a uniform magnetic field $\vec{B}$. So the particle will experience a force $F = qvB$ of constant magnitude directed perpendicular to $\vec{v}$. Under such a force the particle will move in UCM. The magnetic force will serve as centripetal force:

$$qvB = \frac{mv^2}{R} \quad (\text{VI.9}),$$

where R is the radius of the circle. Expression (VI.9) gives: $R = \frac{mv}{qB}$

The radius of the orbit is directly proportional to the linear momentum of the particle and inversely proportional to the magnetic field strength. The period of the orbit is:

$$T = \frac{2\pi R}{v} = \frac{2\pi m}{qB} \quad (\text{VI.10})$$

And the frequency $f = \frac{1}{T}$ is

$$f_c = \frac{qB}{2\pi m} \quad (\text{VI.11})$$

Because of its importance in the operation of particle accelerator called the "Cyclotron", f_c is called the cyclotron frequency. Expressions (VI.10) and (VI.11) lead to two important conclusions:

- (i) Both the period and the frequency are independent of the speed of the particle
- (ii) All particles with the same charge-to-mass ratio, $\frac{q}{m}$ have the same period and cyclotron frequency

This can be used in separating particles with different parameters. An example of such applications is the mass-spectrometer.

Mass spectrometer

A mass spectrometer is a device that separates charged particles, usually ions according to their charge-to-mass ratio. If the charges are the same, the instruments can be used to measure the mass of the ions. In particular it can be used to separate isotopes that are atoms with the same charge but different mass of the nucleus.

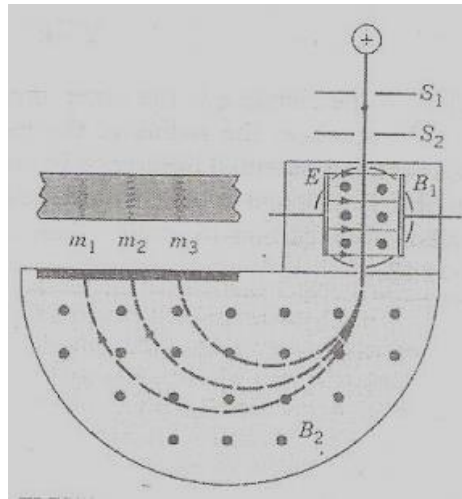


Fig.VI.7. Principle of mass-spectrometer

VI.8. COMBINED ELECTRIC AND MAGNETIC FIELDS

When a particle is subjected to both electric and magnetic fields in the same region, the total force on it is:

$$\vec{F} = q(\vec{E} + [\vec{v} \times \vec{B}]) \quad (\text{VI.12})$$

This is called the **Lorentz* force**. The path of a particle in a uniform magnetic field is a helix, and the path in a uniform electric field is a parabola. When a particle is subject to both electric and magnetic fields, the motion is in a general case quite complex. However the special cases in which the fields are either parallel or perpendicular to each other are useful and simple to analyse

***Hendrik Anton Lorentz (18 July 1853 – 4 February 1928)** was a Dutch physicist who shared the 1902 Nobel Prize in Physics with Pieter Zeeman for the discovery and theoretical explanation of the Zeeman effect. He also derived the transformation equations subsequently used by Albert Einstein to describe space and time.

Summary

The magnetic force on a charge q with velocity $\vec{v}$ in a magnetic field $\vec{B}$ is :

$$\vec{F} = q[\vec{v} \times \vec{B}]$$

The magnitude of the force is: $\vec{F} = qv.b \sin \phi$ and its direction is always perpendicular to both $\vec{v}$ and $\vec{B}$. A magnetic force can accelerate a particle but cannot do work on the particle and therefore can change the energy of the particle.

The magnetic force on straight wire of length ℓ carrying a current I in a uniform magnetic field $\vec{B}$ is

$$\vec{F} = I[\ell \times \vec{B}]$$

The magnetic dipole moment of a coil of N turns of cross-sectional area A and carrying a current I is : $\mu = NIA\vec{n}$

,where $\vec{n}$ the unit vectro perpednicular the the plane of the coil. When a coil of magnet moment μ is placed in a unifrom field $\vec{B}$ it experiances a torque:

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad \text{that tend to align } \mu \text{ along } \vec{B}$$

When a particle of charge q and mass m moves with velocity $\vec{v}$ in a plane perpendicular to a uniform field, the particle moves in a circular path. From the Newton's second law: $qvB = \frac{mv^2}{R}$

The period $T = \frac{2\pi R}{v}$ and the cyclotron frequency f_c are given by: $f_c = \frac{1}{T} = \frac{qB}{2\pi m}$

Note that these quantities are not dependent on v or R

A particle subject to both electric field and a magnetic field experiences the Lorentz force: $\vec{F} = q(\vec{E} + [\vec{v} \times \vec{B}])$

Self-test VI

1. Can a magnetic field be used to speed up a charged particle. Explain why or why not?
2. In the equation $\vec{F} = q[\vec{v} \times \vec{B}]$, which pairs of quantities are always perpendicular? Which one can be at any angle?
3. How does the radius of the path of a charged particle in a magnetic field depend on its kinetic energy?
4. A beam of electrons passes undeflected through a region. What possible conclusions could be drawn regarding the existence of electric and magnetic fields?
5. In a certain region the earth's field is horizontal and points due North. What is the direction of the force on an electron moving: (i) North, (ii) East, (iii) Vertically up?
6. Cosmic rays are high-energy charged particles that bombard the Earth from space. The flux of cosmic ray particles reaching the earth is higher at the Poles than at equator. Why is this?
7. Why does a bar magnet placed close to side of a TV tube distort the picture on the screen?
8. Fig.VI.8. shows the path of an electron that passes through two regions containing uniform magnetic fields of magnitude B_1 and B_2 . Its path in each region is a half-cycle.
 - (i) Which field is stronger?
 - (ii) What are the directions of the two fields?
 - (iii) In which of the regions does the electron spend more time?

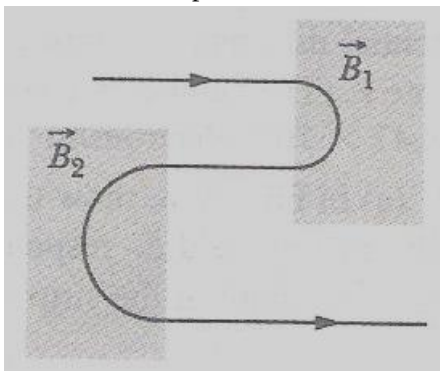


Fig.VI.8. Self-test 8.

Problems VI

1. A proton travelling at 23.0° with respect to the direction of a magnetic field of strength 2.60mT experiences a magnetic force of $6.5 \times 10^{-17}\text{ N}$. Calculate:
 - (i) The proton's speed
 - (ii) Its kinetic energy in electron-volt

2. An electron that has velocity: $\vec{v} = (2.0 \times 10^6)\vec{i} + (3.0 \times 10^6)\vec{j}, \frac{m}{s}$ moves through a magnetic field $\vec{B} = (0.03)\vec{i} - (0.15)\vec{j}, T$
- Find the force on the electron
 - Repeat the your calculations for a proton having the same velocity
3. An electric field of $1.5 \frac{kV}{m}$ and a magnetic field of $0.4T$ act on a moving electron to produce no net force
- Calculate the minimum speed of the electron
 - Draw the vectors $\vec{E}$, $\vec{B}$ and $\vec{v}$
4. A transmission line carries a current of $10^3 A$ from West to East. The Earth's field is horizontal and points due North, and has a magnitude of $0.5G$. What is the force on one metre of the wire?
5. The armature of a motor has 8 square coils of side $10cm$. They are all perpendicular to a radially directed magnetic field of magnitude $0.2T$ as shown in Fig.VI.9. If the current is $10A$ find:
- The torque produced
 - The mechanical power output at $1200rpm$

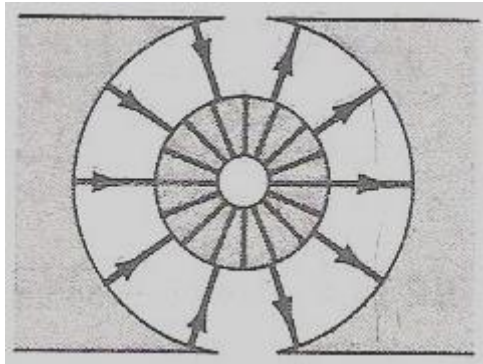


Fig.VI.9. Problem 5

6. A proton is in a circular orbit of radius $20cm$ perpendicular to a $0.8T$ magnetic field. Find:
- Its speed
 - Its period
 - Its kinetic energy
7. An alpha particle of mass $6.7 \times 10^{-27} kg$ and charge $2e$ is accelerated from rest by a potential difference of $17kV$ and enters a uniform magnetic field of $0.6T$ normal to the lines. Find the radius of its path
8. An alpha particle with charge $2e$ has velocity $\vec{v} = (2\vec{i} - \vec{j} + 3\vec{k}) \times 10^6, \frac{m}{s}$ in a uniform field $\vec{B} = (-0.1\vec{i} + 0.2\vec{j} + 0.5\vec{k}) T$. What is the magnetic force on the particle?
9. A wire of length $2m$ carries a $25A$ current in the $-z$ direction which is perpendicular to a uniform magnetic field. The force on the wire is $\vec{F} = (4\vec{i} - 6\vec{j}) \times 10^{-5}, N$. Determine $\vec{B}$
10. Electrons moving at 30° to the lines of a $0.06T$ magnetic field experiences a force of $5.1 \times 10^{-14} N$. What is the kinetic energy of the particles in eV ?

LECTURE VII: SOURCES OF MAGNETIC FIELD

VII.1. LECTURE OUTLINE

In the previous lecture we learnt that a magnetic field can act on moving charges and consequently on conductors carrying current. Now the question is: where the magnetic field comes from? Magnetic fields that do not vary in time are produced by conductors carrying DC currents according to the Biot- Savart law. The simplest case for calculations is the magnetic field created by a long, straight wire with current. We will see also that to get an strong magnetic field such a straight wire is not the best arrangement but rather a solenoid, a wire coiled around its axis. For generality we will learn the Ampere's law that allows us to calculate the magnetic field around a contour containing a certain number of currents. In this lecture, also we will learn about the interaction between conductors carrying currents and the definition of the Ampere as a base SI units. Finally we will learn how permanent magnets are created and magnetic proprieties of matter

Specific objectives

At the end of this lecture, you should be able to:

1. State and apply the Biot-Savart law for simple distribution of currents
2. Calculate the magnetic field created by a long straight wire with current
3. Calculate the force of interaction between two conductors with current
4. Formulate and apply the Ampere's law
5. Compute the magnetic field inside a solenoid with current
6. Describe the magnetic field of the Earth
7. Describe the magnetic proprieties of different material
8. Explain how permanent magnets are created.

VII.2. MAGNETIC FIELDS DUE TO CURRENTS

One way to produce a magnetic field is with moving charges: with a current. We use the same procedure as to calculate the electric field intensity E produced by an arbitrary distribution of stationary electric charges as shown in Fig.VIII.1(a). In Fig.VII.1 (b), a wire of arbitrary shape with current i . What is the magnetic field B at a point P?

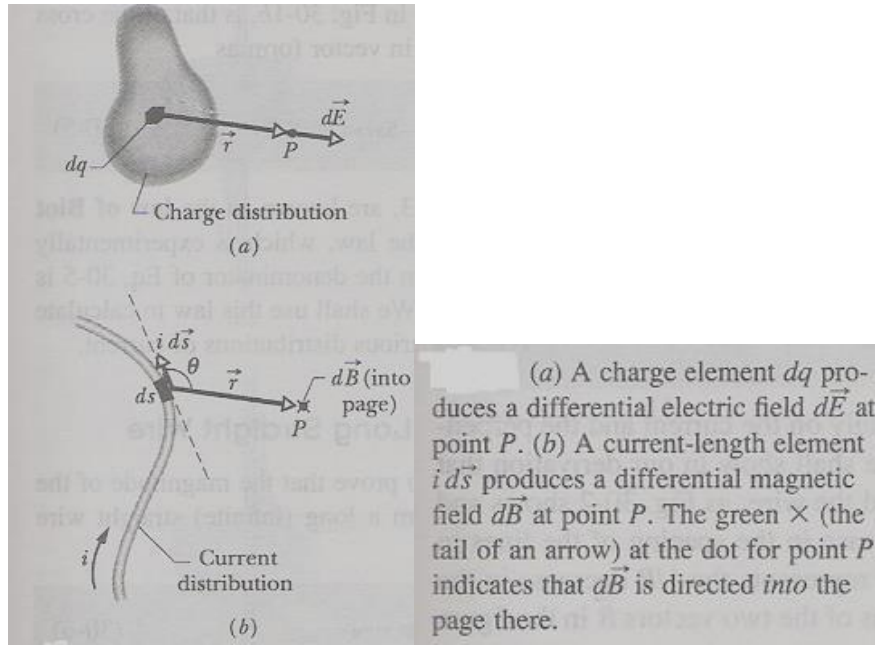


Fig.VIII.1. A distributed electric charge and a distributed current

We divide the wire into differential elements dl . We get a differential current-length element idl . We calculate the magnetic field $d\vec{B}$ produced at P as:

$$dB = \frac{\mu_0}{4\pi} \frac{idl \sin \theta}{r^2} \quad (\text{VII.1}),$$

where θ , angle between the directions of dl and r

μ_0 is a constant called the **permeability of free space** or **magnetic constant**

$$\mu_0 = 4\pi \times 10^{-7} \frac{T.m}{A} \quad (\text{VII.2})$$

$$dB = \frac{\mu_0}{4\pi} \frac{id\vec{l} \times \vec{r}}{r^3} \quad (\text{VII.3})$$

Expression (VII.3) is called the “**Biot-Savart law**”

VIII.2. MAGNETIC FIELD DUE TO A CURRENT IN A LONG STRAIGHT WIRE

We use the Biot-Savart law to calculate the magnitude of the magnetic field $\vec{B}$ produced at a perpendicular distance R from a long (infinite) straight wire carrying a current i , Fig.VII.2 illustrates the configuration of magnetic lines of force and Fig.VII.3 explains the principle of the **right-hand rule**.

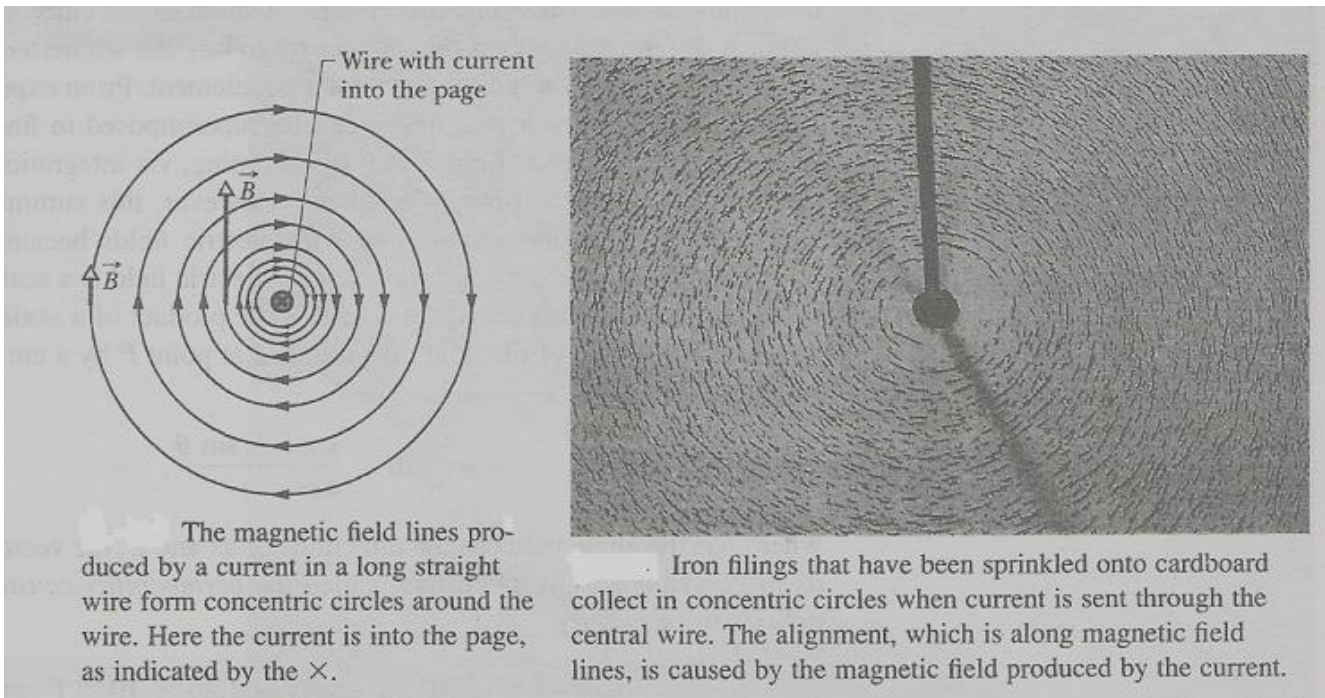


Fig.VII.2. Magnetic field lines produced by a straight wire with current

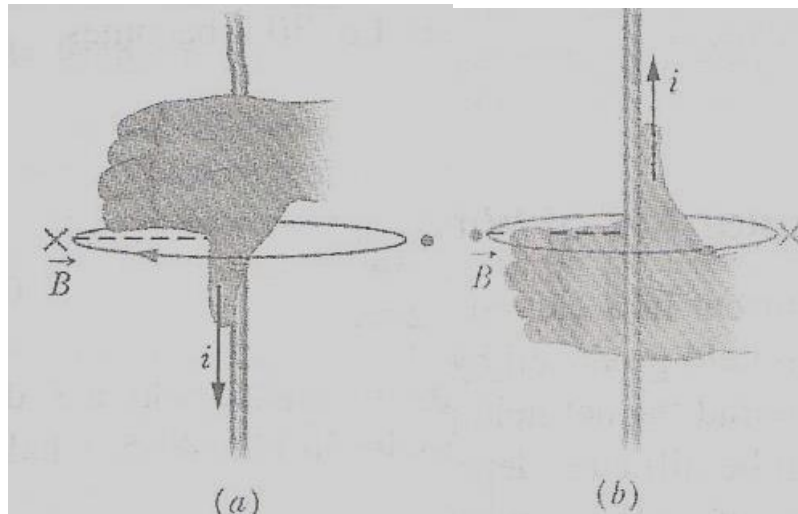


Fig.VII.3 Illustration of the right-hand rule

The magnitude of the differential magnetic field produced at P by the current length-length element ids located a distance r from P is given by (VII.1):

$$dB = \frac{\mu_0}{4\pi} \frac{idl \sin \theta}{r^2} \quad (\text{VII.1}) \quad \rightarrow \quad B = 2 \int_0^\infty dB = \frac{\mu_0 i}{2\pi} \int_0^\infty \frac{\sin \theta dl}{r^2}, \text{ but}$$

$$r = \sqrt{s^2 + R^2} \quad \text{and} \quad \sin \theta = \sin(\pi - \theta) = \frac{R}{\sqrt{s^2 + R^2}}$$

$$\rightarrow B = \frac{\mu_0 i}{2\pi} \int_0^\infty \frac{R ds}{(s^2 + R^2)} = \frac{\mu_0 i}{2\pi} \left[\frac{S}{(s^2 + R^2)^{\frac{1}{2}}} \right]_0^\infty = \frac{\mu_0 i}{2\pi R}$$

$$B = \frac{\mu_0 i}{2\pi R} \quad (\text{VII.4})$$

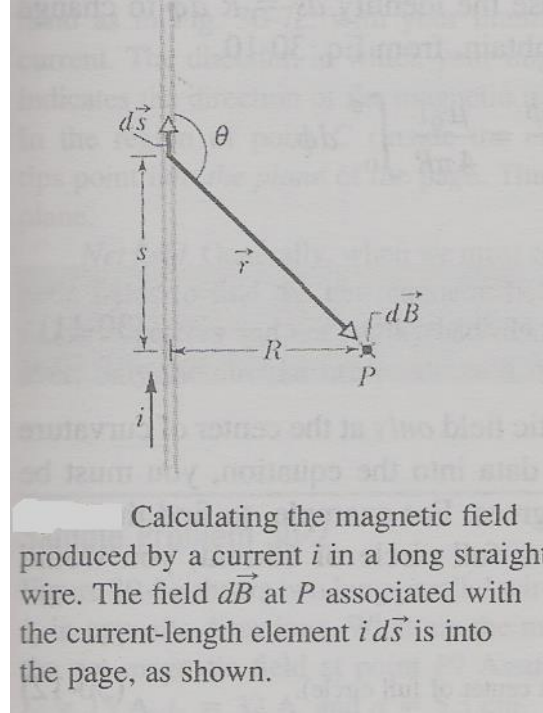


Fig.VIII.4. Magnetic field produced by a current element at a point

VIII.3. MAGNETIC FIELD DUE TO A CURRENT IN CIRCULAR ARC OF WIRE

Fig.VII.5 shows an arc-shaped wire with central angle ϕ , radius R and centre C carrying current i . At C each current-length element ids of the wire produces a magnetic field of magnitude dB given by (VII.1). According to Fig.VII.5 no matter where the element is located on the wire, the angle θ between the vectors ds and r is $\frac{\pi}{2}$ and $r = R$

Substituting this in (VII.1), we get:

$$dB = \frac{\mu_0}{4\pi} \frac{ids \sin \frac{\pi}{2}}{R^2} = \frac{\mu_0}{4\pi} \frac{ids}{R^2}$$

The field at C due to each current-length element in the arc has this magnitude. Applying the right-hand rule shows that each element has the field in C directed out of the page. Thus the total field C is simply the sum of all the differential fields dB . Considering that $ds = R d\phi$ we have:

$$B = \int dB = \int_0^\phi \frac{\mu_0}{4\pi} \frac{iR d\phi}{R^2} = \frac{\mu_0 i}{4\pi R} \int_0^\phi d\phi = \frac{\mu_0 i}{4\pi R} \phi$$

$$B = \frac{\mu_0 i}{4\pi R} \phi \quad (\text{VII.5})$$

This expression gives the value of the field only at the centre of curvature of the current arc. For a full circle of current at the centre C the magnetic field is:

$$B = \frac{\mu_0 i}{4\pi R} 2\pi = \frac{\mu_0 i}{2R} \quad (\text{VII.6})$$

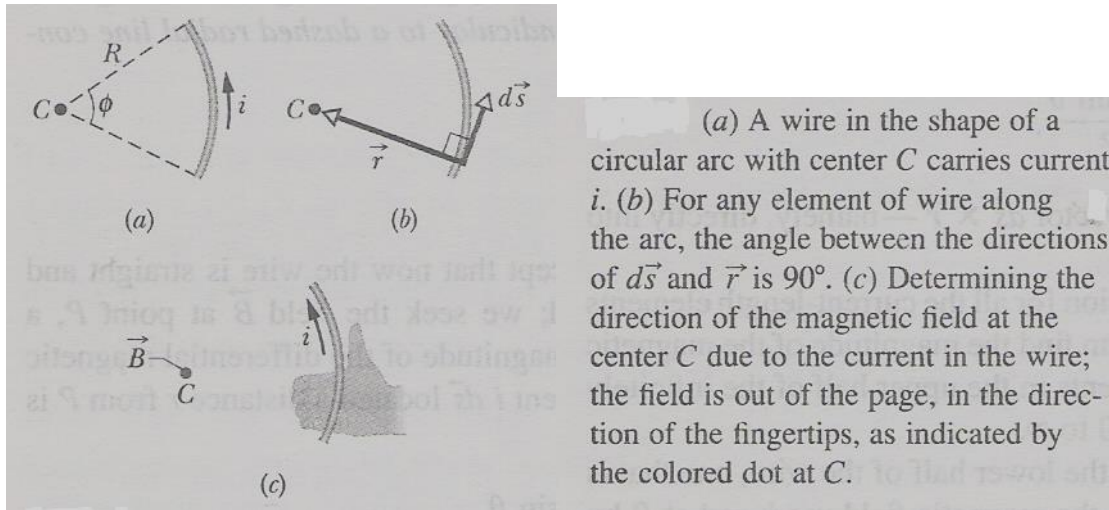


Fig.VIII.5. Magnetic field produced by a arc with current

VII.4. FORCE BETWEEN TWO PARALLEL CURRENTS

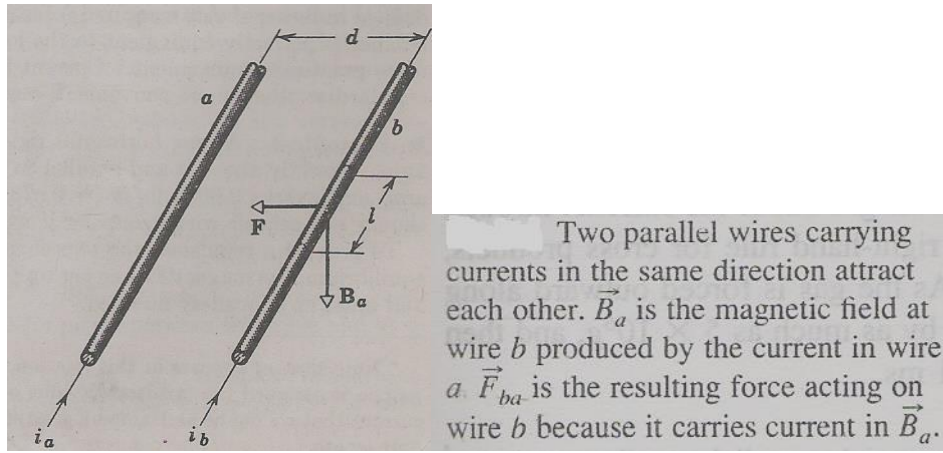
Two long parallel wires carrying currents exert forces on each other. The force on wire **b** in Fig.VII.6 is due to the current in wire **a**. That current produces a magnetic field B_a and it is this magnetic field that actually causes the force acting on **b**.

The magnitude of the field B_a at the site of wire b:

$$B_a = \frac{\mu_0 i}{2\pi d} \quad (\text{VII.7})$$

A right-hand rule shows that the direction of B_a at wire b is down. The force F_{ba} on a length L of wire **b** due to the external magnetic field B_a ;

$$\vec{F}_{ba} = i_b (\vec{L} \times \vec{B}_a) \quad (\text{VII.8})$$



VIII.6. Interaction between two parallel conductors with current

Vectors $\vec{L}$ and $\vec{B}_a$ are perpendicular:

$$F_{ba} = i_b L B_a \sin \frac{\pi}{2} = \frac{\mu_0 L i_a i_b}{2\pi d} \quad (\text{VII.9})$$

The direction of F_{ba} is the direction of the cross product $\vec{L} \times \vec{B}_a$ and according the right-hand rule for the cross product F_{ba} is directed toward wire **a**.

The definition of the Ampere

The force acting between currents in parallel wires is the basis for the definition of the **Ampere** which is one of the seven **base SI units**. This definition that was adopted in 1947 is:

“The Ampere is that constant current which, if maintained in two straight parallel conductors of infinite length, of negligible circular cross section, and placed 1m apart in vacuum would produce on each of these conductors a force of magnitude 2×10^{-7} Newton per metre of length”

VII.5. AMPERE’S LAW

We can find the net magnetic field due to any distribution of currents with the inverse-square law for the differential field dB . If the distribution has some symmetry, the calculations can be easy if we use the Ampere’s law which can be derive from the Biot-Savart law

Ampere’s law is:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} \quad (\text{VII.10})$$

The circle on the integral sign means that the scalar product $\vec{B} \cdot d\vec{s}$ is to be integrated around a closed loop, called **Amperian loop**. The current i_{enc} is the net current encircled by that loop as shown in Fig.VII.7.

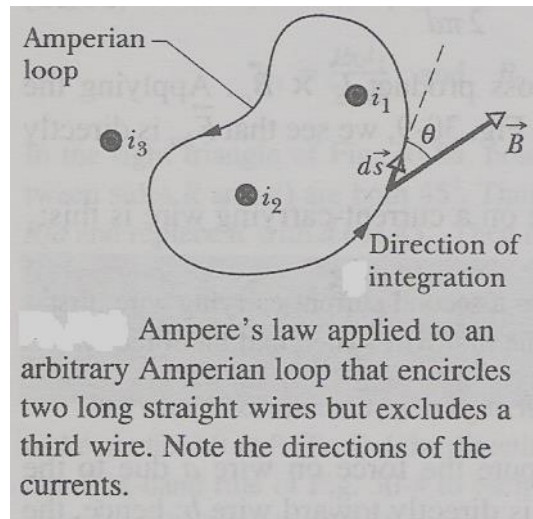


Fig.VII.7. Application of Ampere's law

VII.6. MAGNETIC FIELD DUE TO A CURRENT IN A LONG STRAIGHT WIRE

Fig.VII.8 shows the cross section of a long straight wire of radius R carrying a uniformly distributed i directed out of the page.

The magnetic field B produced by the current has the same magnitude at all points that are the same distance r from the wire: the field has cylindrical symmetry about the wire. We use this symmetry in Ampere's law and we encircle the wire with a concentric circular Amperian loop of radius r as in Fig.VII.8. We note that at any point of the loop B and $d\vec{s}$ are parallel:

$$\oint \vec{B} \cdot d\vec{s} = \oint B \cdot \cos \theta ds = B \oint ds = B(2\pi r)$$

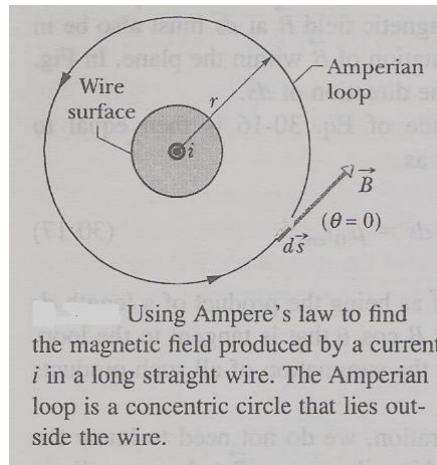


Fig.VII.8. Magnetic field produced by straight wire by Ampere's law

The right side of the integral is $\mu_0 i$: $B(2\pi r) = \mu_0 i$ or

$$B = \frac{\mu_0 i}{2\pi r} \quad (\text{VII.11})$$

Expression (VII.11) is just the same as (VII.4) but it has been derived more easily using the Ampere's law.

VII.6. SOLENOIDS AND TOROIDS

VII.6.1. Magnetic field of a solenoid

A **solenoid** is a long, tightly wound helical coil of wire carrying a current. The length is assumed to be much greater than the diameter as shown in Fig.VII.9.

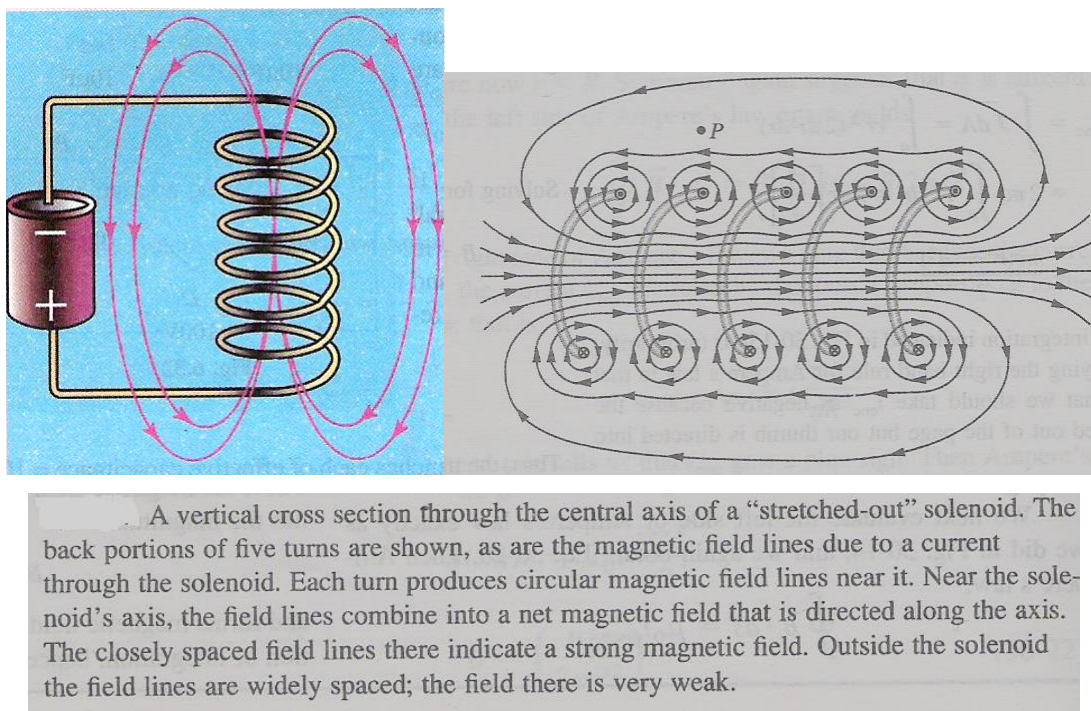


Fig.VII.9. A solenoid

The magnetic field is the vector sum of the fields produced by individual turns or loops that make up the solenoid. The field inside the solenoid is uniform and strong. The external field is weak and negligible.

Let us apply the Ampere's law to the loop **abcd** in Fig.VII.10

$$\oint B \cdot ds = \int_a^b B \cdot ds + \int_b^c B \cdot ds + \int_c^d B \cdot ds + \int_d^a B \cdot ds$$

The first integral is Bh ; the second and the fourth give zero because B and ds are perpendicular to each other. The third gives also zero because the field outside the solenoid is almost zero.

Let n be the number of turns per unit length of the solenoid, then the loop encloses nh turns and:

$$i_{enc} = i(nh)$$

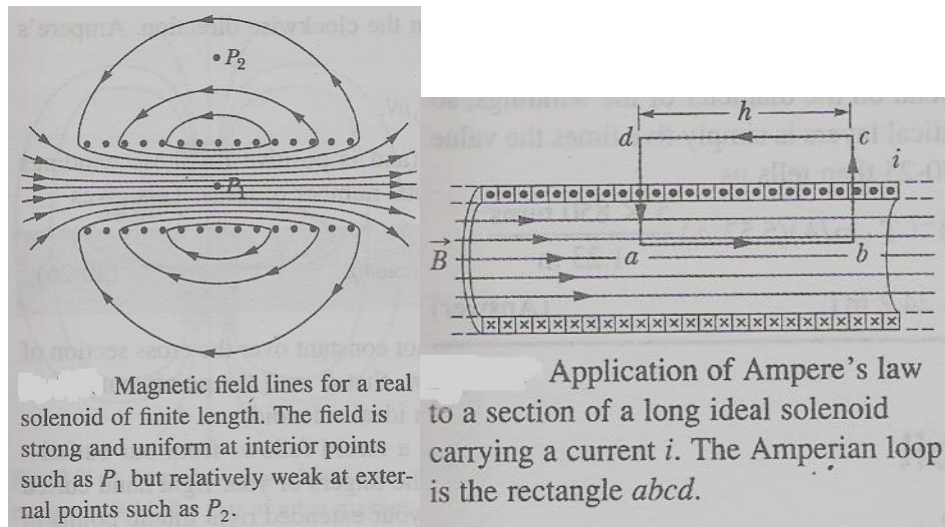


Fig.VIII.10. Calculating the magnetic field inside a solenoid

Ampere's law gives: $Bh = \mu_0 nh$ $\rightarrow$

$$B = \mu_0 ni \quad (VII.12),$$

where n is the number of turns per unit length of the solenoid: the loop encloses nh turns.

We see that the magnetic field magnitude B within a solenoid does not depend on the diameter or length of the solenoid: B is uniform over the solenoid cross section.

Therefore a **solenoid is a structure that helps to create a uniform magnetic field in space just like a plane-plate capacitor helps to create a uniform electric field.**

VII.6.2. Magnetic field of a toroid

Fig.VII.9 shows a toroid, which can be described as a solenoid bent into the shape of a hollow doughnut. The field lines are closed inside the toroid.

The total number of turns:

$$B = \frac{\mu_0 Ni}{2\pi r} \quad (VII.13)$$

In contrast to the situation for the solenoid, B is not constant over the cross section of a toroid. It is easy to show with the Ampere's law that $B = 0$ for all points outside an ideal toroid.

We can take a concentric circle of radius r as an Amperian loop and traverse it in the clockwise direction, the Ampere's law gives: $(B)(2\pi r) = \mu_0 Ni$,

where i is the current in the windings and is positive for those windings enclosed by the Amperian loop and N

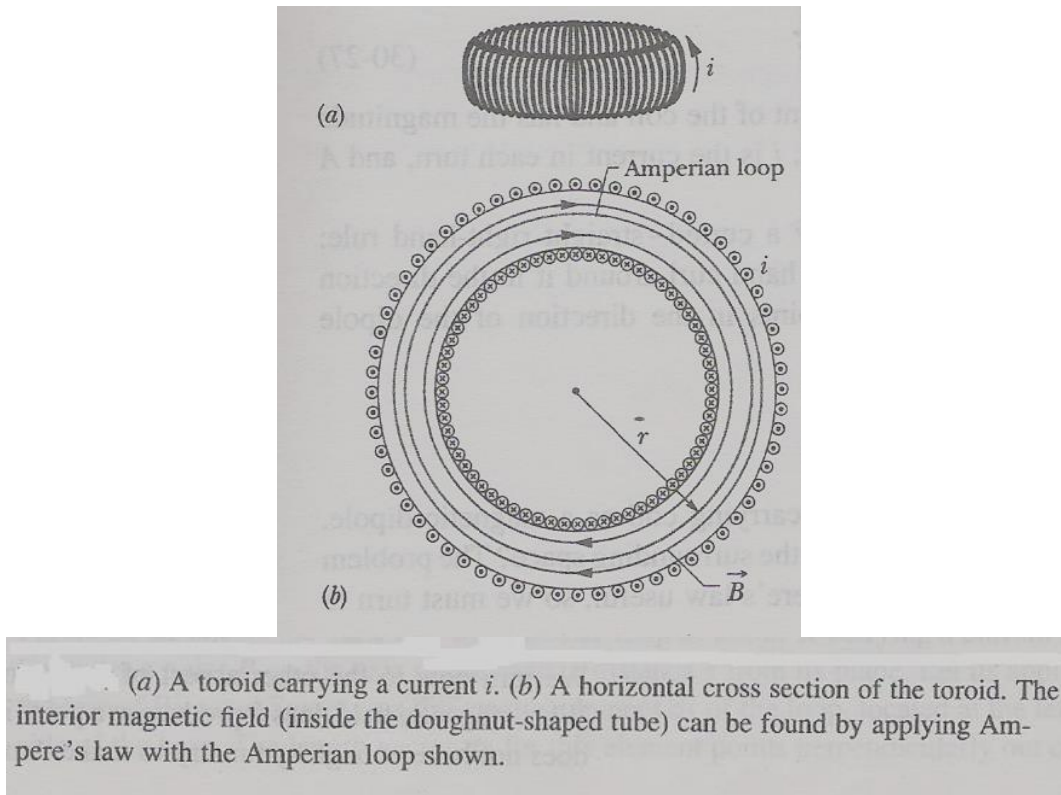


Fig.VII.11. A toroid with current

VII.7. CURRENT CARRYING COIL AS A MAGNETIC DIPOLE

We have seen that a coil carrying current behaves as a magnetic dipole in that, if we place it in an external magnetic field B , a torque τ :

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (\text{VII.14})$$

acts on it. Here $\vec{\mu}$ is the magnetic dipole moment of the coil:

$$\vec{\mu} = NiA \quad (\text{VII.15}),$$

where N , the number of turn or loops, i , the current in the coil, A , the area enclosed by each turn

The direction of $\vec{\mu}$ is given by a curled-straight right-hand rule: Grasp the coil so that the fingers of your right hand curl around it in the direction of the current, your extended thumb then points in the direction of the dipole moment $\vec{\mu}$

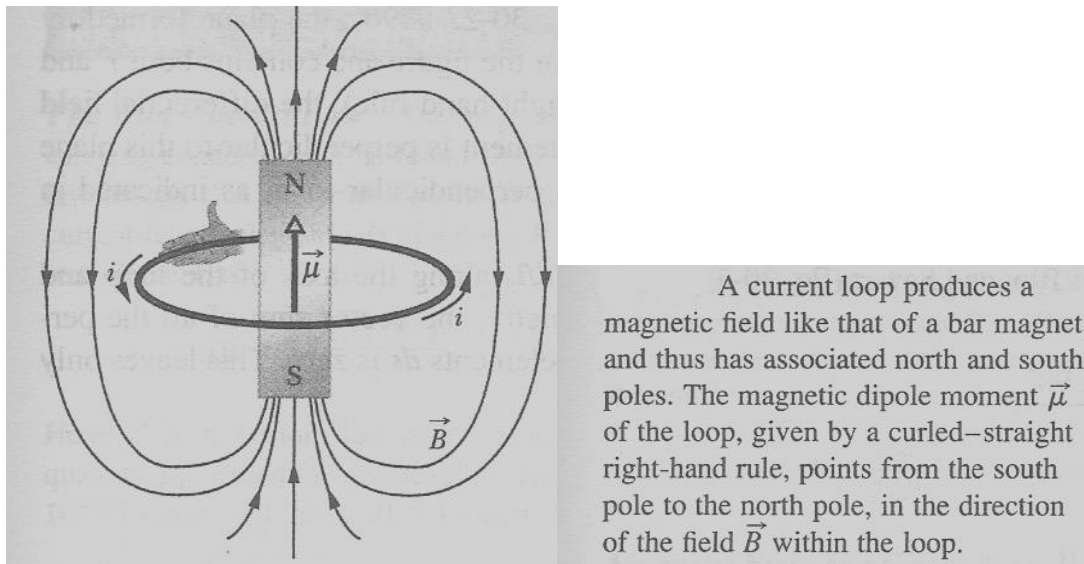


Fig.VII.12. A magnetic dipole

VII.8. MAGNETISM OF MATTER

VII.8.1. Magnets

The first known magnets were “**lodestones**” which are stones that have been magnetized naturally. Ancient Greeks and Chinese were amused by the stones’ ability to attract metal over a short distance. Only much later they learnt to use lodestones (and artificially magnetized pieces of iron) in compasses to determine direction.

Today magnets and magnetic materials are ubiquitous in VCRs, audiocassettes, ATM and credit cards, hard disk drives...

The magnetic properties of materials can be traced back to their atoms and electrons.

If we take a magnet and we break it into two as in Fig.VII.13, we always have a north pole where the magnetic field lines start and the south pole where they sink. Therefore the simplest magnetic structure that exists is a magnetic dipole. Magnetic monopoles do not exist (as far as we know)

VII.8.2. Gauss’s law for magnetic field

Gauss’s law for magnetic fields is a formal manner of saying that magnetic monopoles do not exist. The net magnetic flux Φ_B through any closed Gaussian surface is zero

$$\Phi_B = \oint_S \vec{B} \cdot \vec{A} = 0 \quad (\text{VII.16})$$

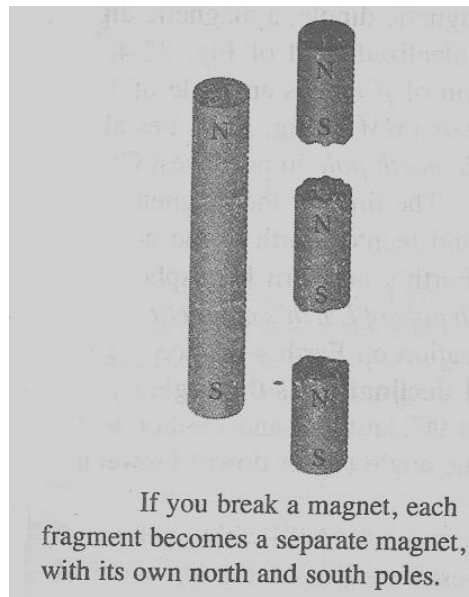


Fig.VII.13. A magnet cut into two parts produces two magnets

VII.8.3.Magnetism of Earth

Earth is a huge magnet: or points near Earth's surface its magnetic field can be approximated as the field of a huge bar magnet- a magnetic dipole- that straddles the centre of the planet. The **magnetic dipole moment** is:

$$\mu = 8.0 \times 10^{22} \frac{J}{T} \quad (\text{VII.17})$$

The direction of $\vec{\mu}$ makes an angle of 11.5° with the rotation axis RR of Earth in Fig.VII.14. The dipole axis MM lies along $\vec{\mu}$ and intersects the Earth's surface at the geomagnetic North Pole in northwest far from Greenland and the geomagnetic South Pole in Antarctica.

The lines of the magnetic field B generally emerge in the southern hemisphere and re-enter Earth in the northern hemisphere

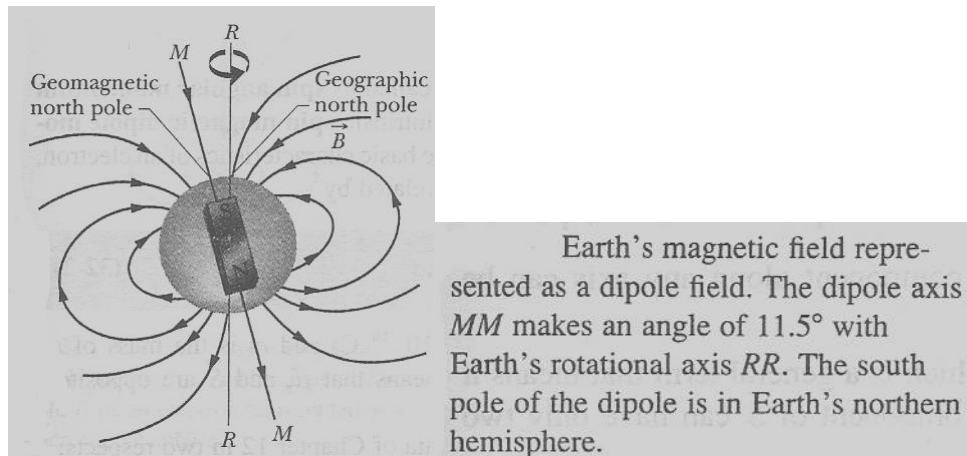


Fig.VII.14. Earth as a huge magnet

Therefore the magnetic pole in the Earth's northern hemisphere and known as "**north magnetic pole**" is really the "**south pole of Earth's magnetic dipole**".

The direction of the magnetic field at any location on Earth's surface is commonly specified in terms of two angles:

- (i) The **field declination**: the angle(left or right) between geographic north and the horizontal component of the field
- (ii) The **field inclination**: up or down between a horizontal plane and the field's direction

It is worth to notice that the field observe at any location on the surface of Earth varies with time, by measurable amounts over a period of a few years and by substantial amounts over centuries. For example between 1580 and 1820 the direction indicated by compass needles in London changed by 35° . In addition research reveals that Earth's field has reversed its polarity (directions of the North Pole and South Pole) about every million. The reason for these reversals is not known. In fact the mechanism that produces Earth's magnetic field is only vaguely understood.

VII.9. MAGNETIC MATERIALS

Each electron in an atom has an orbital magnetic dipole moment and a spin magnetic dipole moment that combine vectorially. The resultant of these two vectors combine vectorially with similar resultants for all other electrons in the atom, and the resultant for each atom combines with those for all the other atoms in a sample of a material.

If the combination of all these dipole moments produces a net magnetic field, then the material is magnetic.

There are three general types of magnetism: Diamagnetism, paramagnetism and ferromagnetism.

1. **Diamagnetism** is exhibited by all common materials but it is so feeble that it is masked if the material also exhibits magnetism of either of the other two types.

In diamagnetism, weak magnetic dipole moments are produced in the atoms of the material when the material is placed in an external magnetic field B_{ext} . The dipole moments and their net fields disappear when B_{ext} is removed

2. **Paramagnetism** is exhibited by materials containing transition elements, rare earth elements and actinide elements.

Each atom of such a material has a permanent resultant magnetic dipole moment but the moments are randomly oriented in the material and as a whole it lacks a net magnetic field. An external magnetic field B_{ext} can partially align the atomic magnetic dipole moments to give a net magnetic field which disappears when the external field is removed.

3. Ferromagnetism is propriety of iron, nickel and certain other elements and alloys of these elements. Some of electrons in these materials have their resultant magnetic dipole moments aligned which produces regions with strong magnetic moments. These regions are called “**domains**”

An external magnetic field B_{ext} can then align the magnetic moments of such domains producing a strong magnetic field in the sample. The field partially persists when B_{ext} is removed. These materials are called **ferromagnetic materials**.

VII.9.1. Ferromagnetism

When we talk about a magnetic material, we have in view generally a ferromagnetic material. Iron, cobalt, nickel, gadolinium, dysprosium and alloys containing these elements exhibit ferromagnetism because of a quantum physical effect called “**exchange coupling**” in which the electron spins of one atom interact with those of neighbouring atoms. The result is alignment of the magnetic dipole moments of the atom. If temperature of a ferromagnetic material is raised above a certain critical value called the “**Curie temperature**” the exchange coupling ceases to be effective. Most of such materials become paramagnetic.

The Curie temperature for iron is $1043K = 770^{\circ}C$

The magnetization of a ferromagnetic material such as iron can be studied with an arrangement called a Rowland ring illustrated in Fig.VII.15.

The material is formed into a thin toroidal core of circular cross section. A primary coil P having n turns per unit length is wrapped around the core and carries current i_p . If the iron core were not present, the magnitude of the magnetic field inside the core would be: $B_0 = \mu_0 i_p n$

When the iron core is present, the magnetic field B inside the core is greater than B_0 by a large amount:

$$B = B_0 + B_M$$

where B_M is the magnitude of the magnetic field contributed by the iron core

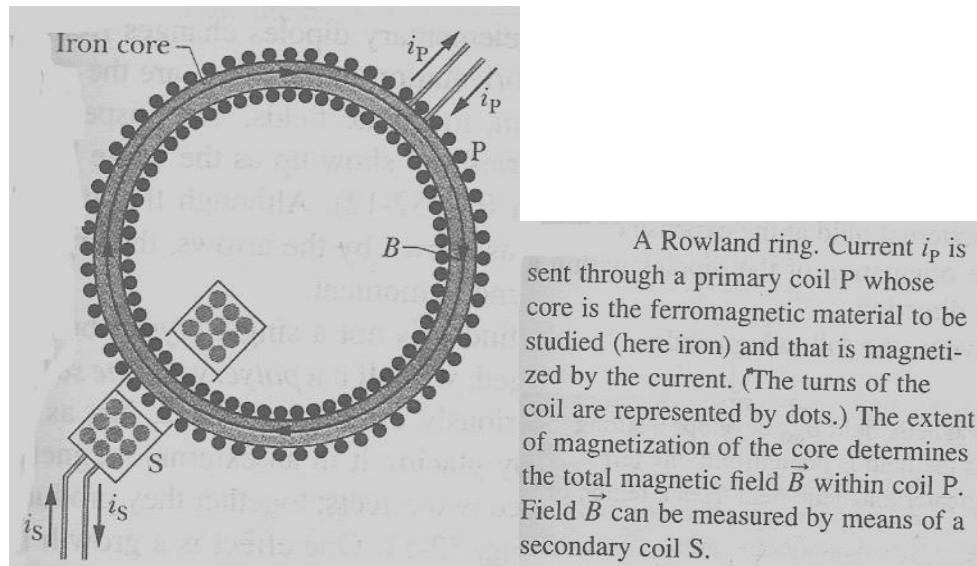


Fig.VII.15. Rowland ring

To determine B_M we use a secondary coil S to measure B and B_0 .

Fig.VII.16 shows the magnetization curve for a ferromagnetic material in a Rowland ring: the ratio

$\frac{B_M}{B_{M,\max}}$ where $B_{M,\max}$ is the maximum possible value of B_M , corresponding to saturation, is plotted versus B_0

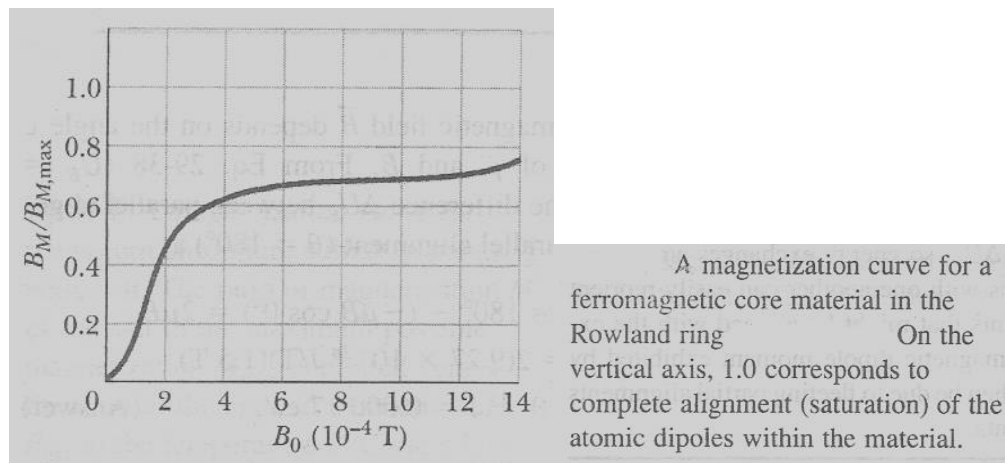


Fig.VII.16. Magnetization curve of a ferromagnetic material

Hysteresis

Magnetization curves for ferromagnetic materials are not retraced as we increase and then decrease the external magnetic field B_0 . The curve in Fig.VII.17 is a plot of B_M versus B_0 during a

Rowland ring experiment

The lack of retraceability is called **hysteresis** and the curve $bcdeb$ is the **hysteresis loop**. Note that at points c and e the core is magnetized even though there is no current in the toroid winding. This is the familiar phenomenon of **permanent magnetism** which is understandable thanks to domains theory. The shape of the hysteresis loop for a given material dictates its applications. Material with flat and large loops are better for construction of permanent magnets, materials with thin and non saturated loops are used in devices for AC current such as transformers.

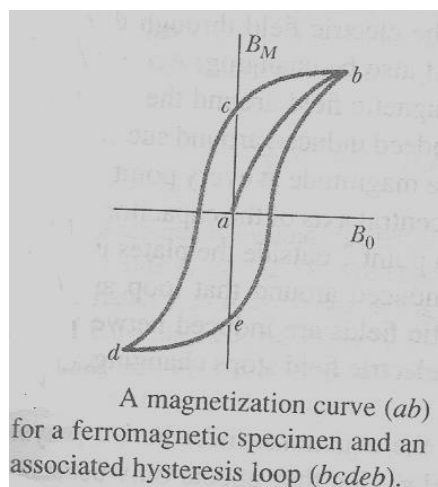


Fig.VII.17. Hysteresis loop

Self-test VII

1. Two long and straight wires current-carrying wires are perpendicular to each other. Describe the forces they exert on each other
2. A weight suspended by a spring ensures that adjacent springs do not touch. What happens when a current passes through the spring?
3. From considerations of symmetry alone, what patterns are possible for the field lines due to an infinite straight wire?
4. A metal tube carries a current along its length. What can you say about the magnitude field within the cavity of the tubes, if the cross section is (i) circular (ii) square?
5. Fig.VII.18 shows cross sections of two long straight wires: the left-hand wire carries current i_1 directly out of the page.. If the net magnetic field due to the two currents is to be zero at point P,
 - (i) Should the direction of current i_2 in the right-hand wire be directly into or out the page and
 - (ii) Should i_2 be greater than, less, or equal to i_1 ?

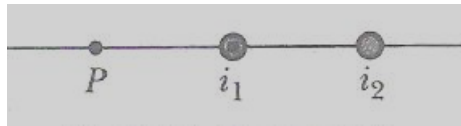


Fig.VII.18. Self-test 5

6. What are the dimensions of the quantity $\frac{1}{\sqrt{\epsilon_0 \mu_0}}$? What is its numerical value?

Practical activity VII

Problems VII

1. A long, straight wire and a rectangular loop lie in the same plane, as shown in Fig.VII.19. The dimensions and currents are indicated. Find net force on the loop

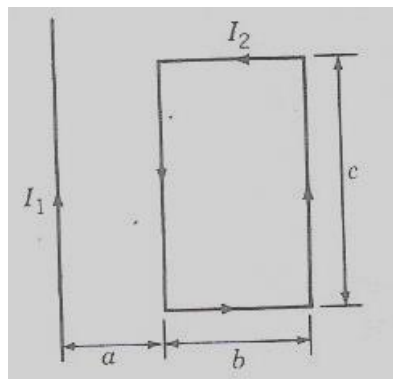


Fig.VII.19. Problem 1

2. Two long, straight wires carry currents in the directions shown in Fig.VII.20
- Find the magnetic field strength at point P
 - What force would 1m of a third wire, carrying a 3A current out of page, if placed at P?

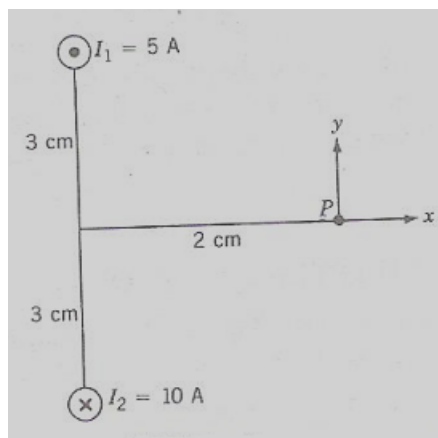


Fig.VII.20 Problem 2

3. A DC power line 20m above the ground carries a 600A current due North. If the horizontal component of the Earth's field is 0.5G due North, in what direction does a compass needle, on the ground below the line, point?

4. A straight infinite current-carrying wire is bent into the form shown in Fig.VII.21. The curve is a semicircle of radius a . What is the magnetic field at point P?

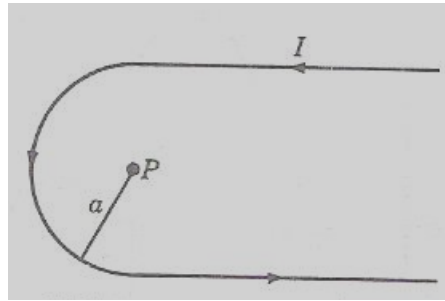


Fig.VII.21. Problem 4

5. A current loop consists of two concentric semicircles as shown in Fig.VII.22. What is the field strength at the centre of the semicircles

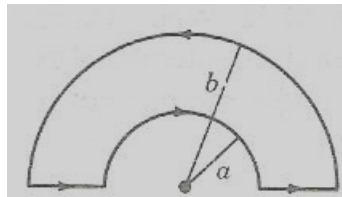


Fig.VII.22. Problem 5

6. A metal tube has an inner radius a and outer radius b . It carries a current I uniformly spread over its cross sectional area. Find the magnetic field for the regions: (i) $r < a$ (ii) $a < r < b$ and (iii) $r > b$
7. A long, straight wire of radius 2mm carries a 12A current uniformly distributed throughout its cross section. At what points, both inside and outside the wire, is the field strength 25% the value at the surface of the wire?
8. Two infinite parallel wires are separated by 2.5cm and carry currents of 10A and 12A in the same direction. What is the force per unit length on each other?
9. A flat circular coil of radius 4cm carries a 1.3A current and produces a magnetic field of 4.9G at its centre. How many turns does it have?

PART IV TIME VARYING ELECTROMAGNETIC FIELDS

LECTURE VIII: TIME VARYING ELECTROMAGNETIC FIELDS

VIII.1. LECTURE OUTLINE

In this lecture, we will learn the Faraday's law of electromagnetic induction that basically says that a time varying magnetic field produces an electromotive force, emf. This is the principle of electrical power generation all over the world. It also encompasses other basic utilities like transformers and other electric devices. The production of an emf is equivalent to the production of an electric field. From the theoretical point view, the culmination of the theory introduced by Faraday is the system of equations put together by Maxwell* and thus called "**Maxwell's equations**". The solution of these equations predicts inter alia the possibility of propagation of electromagnetic waves in space, including free space.

***James Clerk Maxwell** FRS FRSE (13 June 1831 – 5 November 1879) was a *Scottish theoretical physicist*. His most prominent achievement was formulating a set of equations, called "**Maxwell's equations**" that united previously unrelated observations, experiments, and equations of *electricity, magnetism* and *optics* into a consistent theory. His theory of *classical electromagnetism* demonstrates that electricity, magnetism and light are all manifestations of the same phenomenon, namely the *electromagnetic field*. Maxwell's achievements concerning electromagnetism have been called the "**second great unification in physics**", after the first one realised by *Isaac Newton*.

Specific objectives

1. Describe the experiments that helped Faraday to discover the law of electromagnetic induction
2. Define the magnetic flux
3. State and apply Faraday's of electromagnetic induction
4. Describe the principle of an electric generator
5. Explain the effect of self-induction and mutual induction
6. Qualitatively and quantitatively le principle of a transformer and its importance in power transmission

VIII.2. FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION: EXPERIMENTS

VIII.2.1 Two symmetric situations

In the previous section we have seen that if we put a closed conducting loop in a magnetic field and then send current through the loop, then forces due to the magnetic field create a torque to turn the loop:

$$\text{Current loop} + \text{magnetic field} \rightarrow \text{torque} \quad (\text{VIII.1})$$

Suppose that instead, with the current off we turn the loop by hand. Will a current appear in the loop?

$$\text{Torque} + \text{magnetic field} \rightarrow \text{current?} \quad (\text{VIII.2})$$

The answer is yes! The situations of equations (VIII.1) and (VIII.2) are symmetric. Equation (VIII.1) is the principle of *electric motor* and equation (VIII.2) is the *Faraday's law of electromagnetic induction* and the principle of *electric generator*.

VIII.3.2 Two experiments

First experiment

Fig.VIII.1 shows a conducting loop connected to a sensitive galvanometer. Since there is no emf source there is no current in the circuit. However if we move a bar magnet toward the loop, a current suddenly appears in the circuit. The current disappears when the magnet stops. If we then move the magnet away, a current again appears but now in the opposite direction

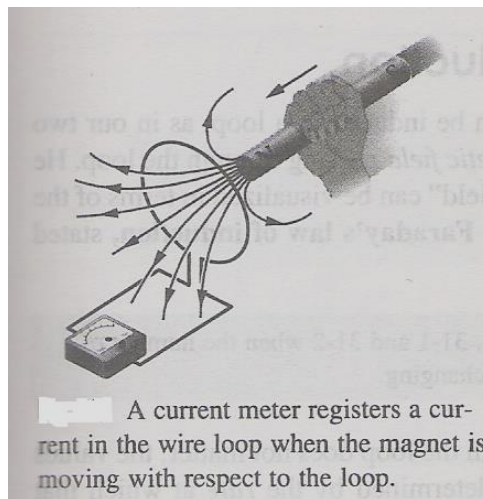


Fig.VIII.1. A moving magnet creates current in a loop

If we perform the experiment many times in different ways we will notice that:

1. A current appears only if there is a relative motion between the loop and the magnet
2. Faster motion produces greater current
3. If moving the magnet's north pole toward the loop causes a clockwise current then moving north pole away causes anticlockwise current and vice-versa for the south pole

The current produced in the loop is called an **induced current**. The work done per unit charge to produce that current is called **induced emf**

This process of producing current and emf is called **electromagnetic induction**

Second experiment

In Fig.VIII.2 we have two conducting loops close to each other but not touching. If we close switch S, to turn on a current in the right-hand loop, the meter in the left-hand suddenly and briefly deflects. If we open the switch the meter deflects again but in the opposite direction

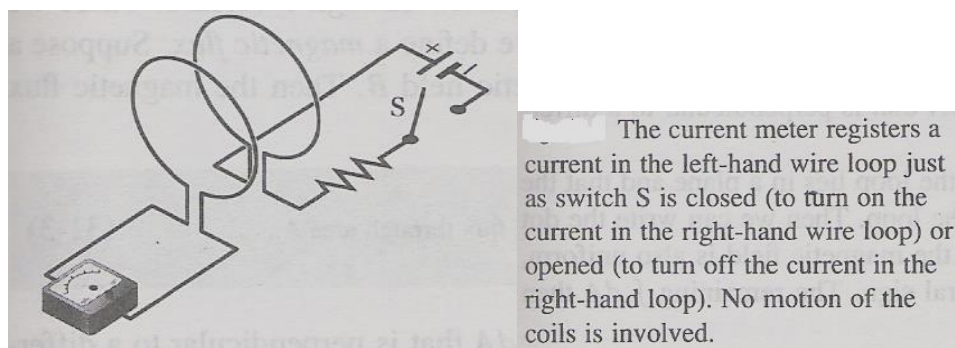


Fig.VIII.2. A varying current in one loop induces current in a close loop

We see that:

1. We get an induced current and emf only when the current in the right-hand loop is changing
2. The induced emf and induced current in these experiments are caused by “**something**” **changing**

Faraday after 12-year long experimentation knew that ”something”

VIII.3. FARADAY’S LAW OF ELECTROMAGNETIC INDUCTION: THEORY

Faraday realized that an emf and a current can be induced in a loop by changing the amount of magnetic field passing through the loop. He further realized that the amount of magnetic field can be visualized in terms of magnetic field lines passing through the loop.

Faraday’s law of electromagnetic induction can be stated:

“An emf is induced in the loop when the number of magnetic field lines that pass through the loop is changing”

A quantitative treatment

To calculate the amount of magnetic field lines passing through the loop we introduce a quantity called ”**magnetic field flux**”.

Suppose a loop enclosing an area A in placed in a magnetic field B . Then the magnetic flux through the loop is:

$$\Phi_B = \int \vec{B} \cdot d\vec{A} \quad (\text{VIII.3})$$

$d\vec{A}$ is a vector of magnitude dA that is perpendicular to a differential area dA

The SI unit for magnetic flux is : *Tesla X square metre* called the *Weber, Wb* :

$$1\text{Weber} \equiv 1\text{Wb} = 1\text{T} \cdot \text{m}^2 \quad (\text{VIII.4})$$

The SI unit for magnetic flux was named after **Wilhelm Eduard Weber***, a German physicist

“The magnitude of the emf induced in a conducting loop is equal to the rate at which the magnetic flux Φ_B through that loop varies with time”

$$E_{ind} = -\frac{d\Phi_B}{dt} \quad (\text{VIII.5})$$

Expression (VIII.5) is called **Faraday’s law of electromagnetic induction**

If the magnetic flux through a coil of N turns an induced emf appears in every turn and the total induced emf is:

$$E_{ind} = -N \frac{d\Phi_B}{dt} \quad (\text{VIII.6}),$$

where N is the number of turns in the coil and Φ_B the magnetic flux through each turn

Expression (VIII.6) can be written:

$$E_{ind} = -\frac{d\Psi_B}{dt} \quad (\text{VIII.6a}),$$

where

$$\Psi_B = N\Phi_B \quad (\text{VIII.6b})$$

is called the “**total magnetic linkage**”

*** Wilhelm Eduard Weber (24 October 1804 – 23 June 1891)** was a German physicist and, together with **Carl Friedrich Gauss**, inventor of the first **electromagnetic telegraph**

There are some means by which we can change the magnetic flux through a coil:

1. Change the magnitude of B of the magnetic field within the coil
2. Change the area of the coil or the portion of that area that lies within the magnetic field
3. Change the angle between the direction of B and the area of the coil

The third means has proved to be the most convenient and most of the power generators are based on the rotation of a coil in a magnetic field.

VIII.4. LENZ'S LAW

Soon after Faraday proposed his law of electromagnetic induction, Heinrich Friedrich Lenz* devised a rule to determine the direction of the induced current. This rule is known as **Lenz's law**:

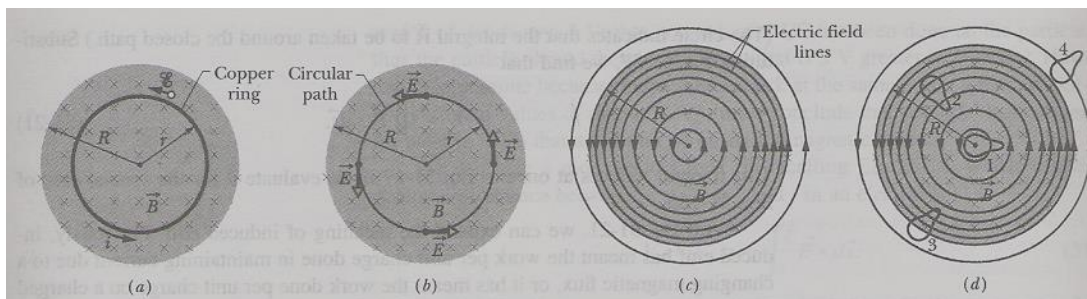
“An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic flux that induces the current”

This explains the sign – in the Faraday's law.

***Heinrich Friedrich Emil Lenz (12 February 1804 – 10 February 1865)** was a **Russian physicist** of **Baltic German ethnicity**. He is most noted for formulating *Lenz's Law* in electrodynamics in 1833. The symbol L , conventionally representing **inductance**, is chosen in his honour.

VIII.5. INDUCED ELECTRIC FIELDS

Let us place a copper ring of radius r in a uniform external magnetic field and change magnetic field B at a steady rate. According to the Faraday's law an induced emf and a current will appear in the ring



(a) If the magnetic field increases at a steady rate, a constant induced current appears, as shown, in the copper ring of radius r . (b) An induced electric field exists even when the ring is removed; the electric field is shown at four points. (c) The complete picture of the induced electric field, displayed as field lines. (d) Four similar closed paths that enclose identical areas. Equal emfs are induced around paths 1 and 2, which lie entirely within the region of changing magnetic field. A smaller emf is induced around path 3, which only partially lies in that region. No emf is induced around path 4, which lies entirely outside the magnetic field.

Fig.VIII.5. Induced electric field

If there is current in the ring there must be an electric field to do work of moving the conduction electrons. This induced electric field produced by a varying magnetic field is as real as an electric field produced by static charges.

→ ***A time-varying magnetic flux produces an electric field***

An important feature of this is that the electric field is induced even if there is no copper ring, in free space for example

A reformulation of Faraday's law

Consider a particle of charge q moving around the circular path. The work W done on it in one revolution by the induced electric field is $E_{ind}q_0$ where E_{ind} is the induced emf.

The work is:

$$\int \vec{F} d\vec{\ell} = q_0 \vec{E}(2\pi r)$$

→ $E_{ind} = 2\pi r \vec{E}$ and more generally the work done on a particle of charge q_0 moving along any closed path is :

$$W = \oint \vec{F} d\vec{\ell} = q_0 \oint \vec{E} d\vec{\ell}$$

Substituting $E_{ind}q_0$ for W we get:

$$\vec{E} = \oint \vec{E} d\vec{\ell} \quad \text{(VIII.11)}$$

By combining equation (VIII.10) and (VIII.6) we can rewrite the Faraday's law of induction:

$$\oint \vec{E} d\vec{\ell} = -\frac{d\Phi_B}{dt} \quad \text{(VIII.12)}$$

This very important equation simply says:

“A time-varying magnetic field produces an electric field, (which generally is also time-varying)”

VIII.6. INDUCTORS AND INDUCTANCE

We have seen that a capacitor can be used to produce a desired electric field. Similarly, an **inductor** is a structure that can be used to produce a desired magnetic field in space.

As a **plane-plate capacitor** is considered as a basic type of capacitor, we can consider a long **solenoid** as a basic type of inductor. Its schematic symbol is:

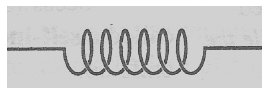


Fig.VIII.6. Schematic symbol of a solenoid or inductor

There is a proportional dependence between the current I and the magnetic linkage $N\Phi_B$ it creates in the central region of a solenoid:

$$\Psi = N\Phi_B = LI \quad (\text{VIII.12})$$

where the proportionality coefficient L is a constant which depends on the size and the shape of the solenoid. It is called “*inductance*” or more formally, the *coefficient of self-induction*

The **SI unit** of inductance is **Henry**, after J. Henry*, an American scientist, discoverer of the law of electromagnetic induction and a contemporary to Faraday:

$$[L] = 1 \frac{T.m^2}{A} \equiv 1 \text{Henry} \equiv 1H \quad (\text{VIII.13})$$

***Joseph Henry (December 17, 1797 – May 13, 1878)** was an American scientist .He was highly regarded during his lifetime. While building **electromagnets**, Henry discovered the electromagnetism **phenomenon of self-induction**. He also discovered **mutual induction** independently of **Michael Faraday** , though Faraday was the first to publish his results. Henry developed the **electromagnet** into a practical device. He invented a precursor to the electric doorbell (specifically a bell that could be rung at a distance via an electric wire, 1831) and electric relay (1835). The **SI unit of inductance**, the **Henry, H**, is named in his honour. Henry's work on the electromagnetic relay was the basis of the practical **electric telegraph**, invented by **Samuel Morse** and **Charles Wheatstone** separately.

Inductance of a solenoid

Consider a length ℓ of a solenoid carrying a current I . The flux linkage is: $\Psi = N\Phi_B = n\ell BA$

where n is the number of turns per unit length of the solenoid

From expression (VII.12) $B = \mu_0 in \rightarrow$

$$L = \frac{N\Phi_B}{i} = \frac{n\ell BA}{i} = \frac{(n\ell)(\mu_0 in)A}{i} = \mu_0 n^2 \ell A \quad (\text{VIII.14})$$

The inductance per unit length for a long solenoid:

$$\frac{L}{\ell} = \mu_0 n^2 A \quad (\text{VIII.15})$$

Inductance, like capacitance depends only on the geometry of the device and usually is not calculated but rather experimentally measured.

VIII.7. SELF-INDUCTION

If two coils or inductors are near each other, a current i in one coil produces a magnetic flux Φ_B through the second coil. If we vary this flux by varying the current, an induced emf appears in the second coil. Experiments show that an induced emf appears in the first coil as well, even in the absence of the second. This can be illustrated by the experiment in Fig.VIII.9.

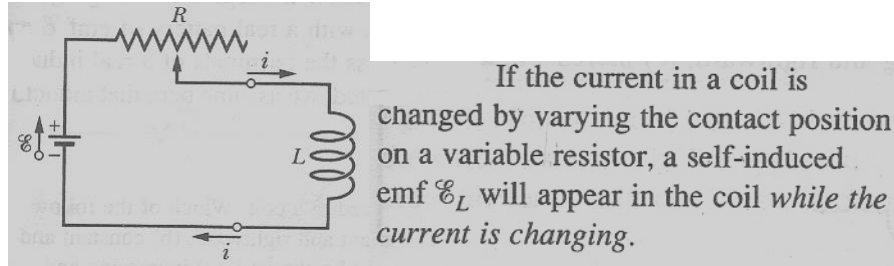


Fig.VIII.7. Demonstration of self-induction

“An induced emf E_{ind} appears in any coil in which the current is varying”

This process is called **“self-induction”** and the emf that appears is called a **“self-induced emf”**. It obeys Faraday’s law of induction just as other induced emfs do.

For any inductor expression (VIII.12) holds: $\Psi = N\Phi_B = Li$

And Faraday’s law says: $E_{ind} = -\frac{d(N\Phi_B)}{dt}$

Combining the two we get:

$$E_{ind} = -L \frac{di}{dt} \quad (\text{VIII.16})$$

Expression (VIII.16) is called **“law of self-induction”** and the direction of emf is found using Lenz’s law:

“Self-induced emf is directed as to resist the change of current in the circuit”

VIII.8. ENERGY STORED IN A MAGNETIC FIELD

We know that a charged capacitor stores electrostatic energy and that energy is stored in the electric field. The same is true for a coil with current and a magnetic energy is stored in the magnetic field. Let us consider a circuit consisting of a source of emf connected to a resistor R and an inductor L in Fig.VIII.8

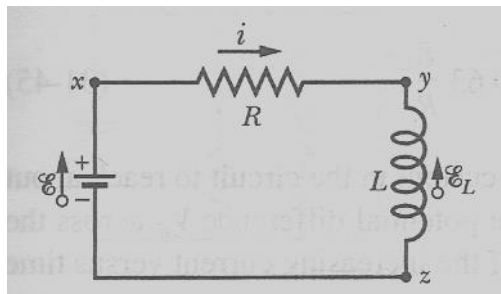


Fig.VIII.8. Circuit consisting of a source of emf, a resistor and inductor

We can write:

$$E = L \frac{di}{dt} + iR \quad (\text{VIII.17})$$

Expression (VIII.17) is the differential equation that describes the growth of the current in the circuit. If we multiply by i both sides:

$$Ei = Li \frac{di}{dt} + i^2 R \quad (\text{VIII.18}),$$

which has the following interpretation in terms of work-energy principle:

1. If a charge dq passes through the battery of emf E in time dt the battery does work Edq on it. The rate at which the battery does work is:

$$P = E \frac{dq}{dt} = Ei$$

2. The rightmost term represents the rate at which energy appears as thermal energy in the resistor we say energy dissipated in the resistor
3. Energy that is delivered to the circuit but is not dissipated as heat, by the conservation of energy principle, must be stored in the magnetic field of the inductor. The middle term represents the rate at which energy is stored in the magnetic field: $\frac{dU_B}{dt}$

$$\begin{aligned} \rightarrow \quad \frac{dU_B}{dt} = Li \frac{di}{dt} \quad \rightarrow \quad dU_B = Lidi \quad \rightarrow \quad \int_0^{U_B} dU_B = \int_0^i Lidi \quad \rightarrow \\ U_B = \frac{1}{2} Li^2 \end{aligned} \quad (\text{VIII.19}),$$

which represents the total energy stored by an inductor L carrying a current i

Note the similarity with equation (IV.10) for energy stored in a charged capacitor: $U_E = \frac{1}{2} CV^2$

Variable i corresponds to V and variable L corresponds to C $\rightarrow$ As for a capacitor the greater the capacitance the more energy a capacitor can store, the higher the inductance of a coil the more energy it can store for the same value of current.

Energy density of a magnetic field

Consider a length ℓ near the middle of a long solenoid of cross-sectional area A carrying current i , the volume is $A\ell$.

The energy U_B stored by the length ℓ of the solenoid lies entirely within the volume because the field is assumed to be zero outside the solenoid. This energy is distributed uniformly as the field.

The energy density per unit volume of the field is: $u_B = \frac{U_B}{A\ell}$

$$\text{Since } U_B = \frac{1}{2} Li^2 \rightarrow u_B = \frac{Li^2}{2A\ell} = \frac{L}{\ell} \frac{i^2}{2A}, \text{ but } \frac{L}{\ell} = \mu_0 n^2 A \rightarrow u_B = \frac{1}{2} \mu_0 n^2 i^2, \text{ but } B = \mu_0 in$$

$$\rightarrow$$

$$u_B = \frac{B^2}{2\mu_0} \quad \text{(VIII.20)}$$

We can compare with energy density of an electric field: $u_E = \frac{1}{2} \epsilon_0 E^2$

VIII.9. MUTUAL INDUCTION

If two coils are close together, a steady current i in one coil will set up a magnetic flux Φ through the other coil (linking the other coil). If we vary i with time, an emf will appear in the second coil. This effect is called “**electromagnetic induction**” and better “**mutual induction**” to distinguish it from “**self-induction**”

Let us consider two circular close-packed coils near each other and sharing a common central axis. There is a steady current i_1 which creates a magnetic field represented by lines of B_1 in Fig.VIII.9

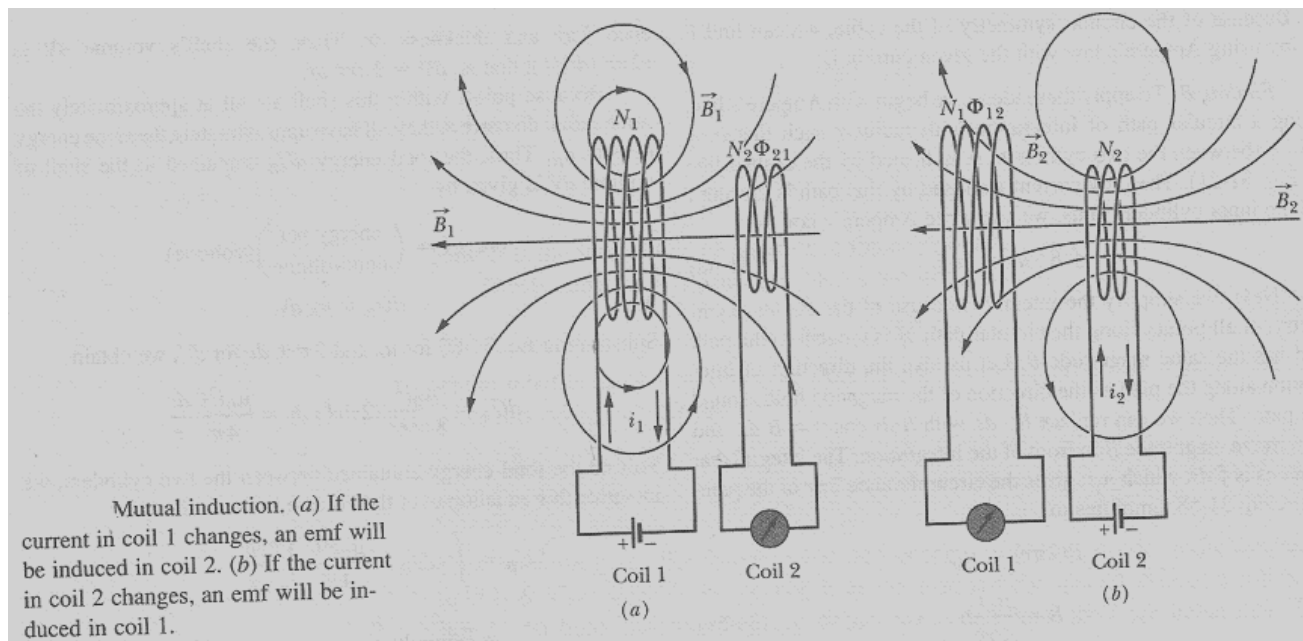


Fig.VIII.9. Illustration of mutual induction

Coil 2 is connected to a sensitive meter but no battery in the circuit. A magnetic flux Φ_{21} links the N_2 turns of coil

2. We define the mutual inductance M_{21} of coil 2 with respect to the current in coil 1:

$$M_{21} = \frac{N_2 \Phi_{21}}{i_1} \quad (\text{VIII.21})$$

or

$$M_{21} i_1 = N_2 \Phi_{21}$$

If i_1 varies with time:

$$M_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}$$

The right side of the equation is just the magnitude of the emf E_{ind2} appearing in coil 2 due to the changing current in coil 1:

$$E_{ind2} = -M_{21} \frac{di_1}{dt} \quad (\text{VIII.22})$$

Let us now interchange the roles of coils 1 and 2: we set up a current i_2 by means of a battery and this produces a

magnetic flux Φ_{12} that links coil 1. If we change i_2 with time: $\rightarrow E_{ind1} = -M_{12} \frac{di_2}{dt}$

Without proof, from the two expressions, we write:

$$M_{21} = M_{12} = M \quad (\text{VIII.23})$$

and

$$E_{ind2} = -M \frac{di_1}{dt}, E_{ind1} = -M \frac{di_2}{dt} \quad (\text{VIII.24})$$

Expressions (VIII.24) are called **laws of mutual induction**

Application of mutual induction: the transformer

The transformer is maybe the most important technical application of the law of mutual induction as such devices are indispensable in modern power transmission and distribution.

The transmission rule leads to a fundamental mismatch between the requirement for efficient high-voltage transmission and the need for safe low-voltage generation and consumption. We need a device with which we can raise (for transmission) and lower (for use) the AC voltage in circuit, keeping the product $I.V = P$ (power) essentially constant.

The transformer is such a device: it has no moving parts, operates by Faraday's law of electromagnetic induction and has no simple DC counterpart

An ideal transformer consists of two coils, with different numbers of turns, wound around an iron core

The iron core is to ensure that the magnetic flux through both windings is the same as it confines the magnetic field lines inside itself

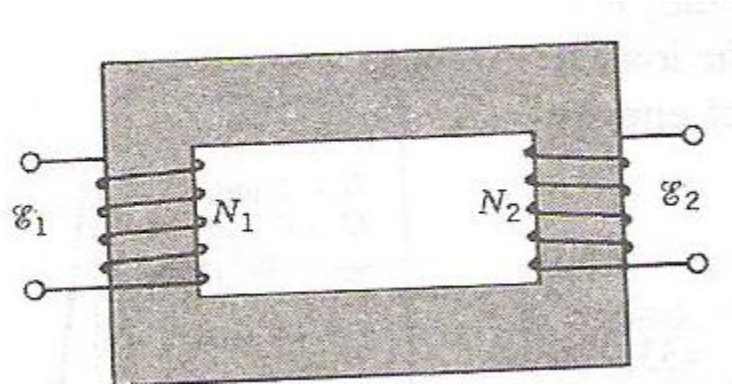


Fig.VIII.13 A simple transformer

The emfs that appear at the primary and the secondary are: $E_{1IND} = -N_1 \frac{d\Phi}{dt}$ and $E_{2IND} = -N_2 \frac{d\Phi}{dt}$,

where Φ is the flux through one turn.

Taking the ratio of these we find:

$$\frac{E_{1IND}}{E_{2IND}} = \frac{V_1}{V_2} = \frac{N_1}{N_2} = n \quad (\text{VIII.25})$$

The quantity $\frac{N_1}{N_2} = n$ is called “**turn ratio**”

Expression (VIII.25) is called the “**law of transformation of voltage**”

If $N_s > N_p \rightarrow V_s > V_p$ - a **step-up transformer**

If $N_s < N_p \rightarrow V_s < V_p$ - a **step-down transformer**

Here the subscript “**p**” stands for “**primary**” and subscript “**s**” stands for “**secondary**”

To calculate I_2 we apply the principle of conservation of energy: energy transferred from the primary is that

dissipated in the secondary: $I_1.V_1 = I_2.V_2 \rightarrow I_2 = I_1 \frac{N_1}{N_2}$ or

$$\frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{n} \quad (\text{VIII.26})$$

Expression (VIII.26) is called the “**law of transformation of current**”

Fig.VIII.14 shows the schematic symbol of a circuit containing a transformer

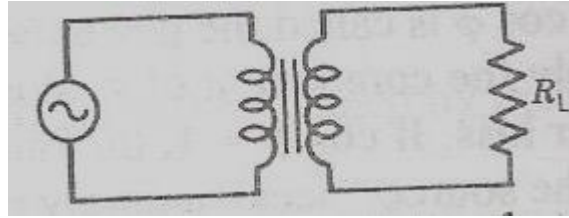


Fig.VIII.14. Schematic representation of a circuit containing a transformer

It is obvious that there are diverse energy losses in a transformer when transferring power from the primary to the secondary circuit, but these losses can be quite low. So the efficiency of an industrial transformer can be as high as 95%

Summary

In a uniform magnetic field, the magnetic flux through a plane surface A is given by:

$$\Phi_B = (\vec{B} \cdot \vec{A})$$

If the area is not flat or the field is not uniform, the flux is given by:

$$\Phi_B = \int_A \vec{B} d\vec{A}$$

Faraday’s law of electromagnetic induction relates the induced emf in a closed loop to the rate of change of flux through it:

$$E_{IND} = -\frac{d\Phi_B}{dt}$$

The negative sign takes into account the direction of E_{IND} which is given by the Lenz’s law: The induced emf opposes the change in magnetic flux that produces it.

The Faraday’s law of electromagnetic induction is the law behind the electric generator and transformer

When two current –carrying coils are close to each other, the magnetic flux in coil 1 has two components:

$$\Phi_1 = \Phi_{11} + \Phi_{12}$$

Where Φ_{11} is the flux in coil 1 due to its own current I_1 , and Φ_{12} is the flux due to the magnetic field produced by current I_2 of coil 2. When the current in the coils change, the net emf in coil 1 is

$$E_{IND1} = E_{IND11} + E_{IND12} = -\frac{d(N_1\Phi_1)}{dt}$$

The self induced emf in coil 1 is given by:

$$E_{IND11} = -L_1 \frac{dI_1}{dt},$$

Where the self-inductance L_1 is found from:

$$N_1\Phi_{11} = L_1 I_1$$

For a solenoid the self-inductance or simply inductance is given by:

$$L = \mu_0 n^2 A \ell,$$

Where n is the number of turns per unit of length of the solenoid: $n = \frac{N}{\ell}$ and A the cross-sectional area of the solenoid or inductor

The emf induced in coil 1 due to changes in I_2 is

$$E_{IND12} = -M \frac{dI_2}{dt},$$

Where the mutual inductance M is found from

$$N_1\Phi_{12} = M I_2$$

Mutual inction is the principle behind the transformer.

A transformer consists of a primary and a secondary coil or winding wrapped around a common soft iron core. The potential differences across the primary and a secondary coil depend on the number of turns in each:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

Such a device transfers power with near ideal efficiency, thus

$$I_1 V_1 = I_2 V_2$$

Energy density in a magnetic field is given by:

$$u_B = \frac{1}{2} \frac{B^2}{\mu_0}$$

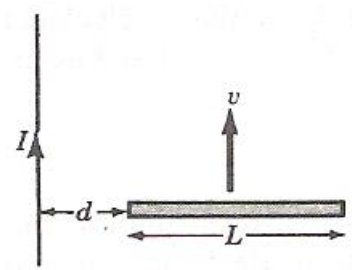
Self-test VIII

- Two circular coils are near each other. How should they be oriented such that their mutual induction is
 - A maximum
 - A minimum?

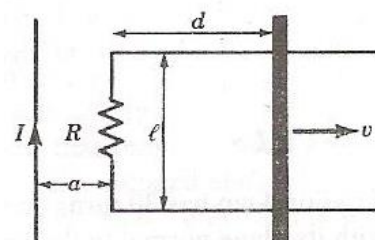
2. Are the relation $L = N \frac{\Phi}{I}$ and $L = -\frac{E_{IND}}{\left(\frac{dI}{dt}\right)}$ equally general? If not, which is preferable as definition of L ?
3. For a solenoid of finite length, is the selfinductance per unit length different at the middle than at the ends? If so, at which point is it greater?
4. Show that the quantity $\frac{1}{\sqrt{LC}}$ has dimension T^{-1}
5. A real inductor has some resistance. Can the potential difference across the terminals of the inductor be
 - (i) Greater than the induced emf
 - (ii) Less than the induced emf
6. What is the effect on the inductance of a solenoid when an iron core is inserted? Does the inductance have a single value?
7. A coil surrounds a long solenoid. The magnetic field due to the solenoid is essentially zero at the position of the coil. Do the coil and solenoid have mutual inductance?
8. Can one have an inductor with resistance? How about a resistor without inductance?
9. Four unmarked wires emerge from a transformer. What steps would you take to determine the turns ratio?
10. Why is electrical power from a power station transmitted at very high potential difference?

Practical activity VIII

1. A long straight wire carries a constant current I . A metal rod of length ℓ moves at velocity v relative to the wire as shown in Fig.VIII.15. What is the potential difference between the ends of the rod?



2. A long straight wire carries a constant current $I = 15A$. A metal rod of length $\ell = 40cm$ moves at constant velocity v on rails of negligible resistance that terminate in a resistor $R = 0.05\Omega$ as shown in Fig.VIII.16. Find



the induced current in the resistor given $a = 1cm$, $d = 5cm$ and $v = 25 \frac{cm}{s}$

3. A square coil has 40 turns of side $4cm$ and a total resistance of 2.5Ω . Its plane is perpendicular to a uniform field that varies in time according to $B(t) = B_0 \exp\left(-\frac{t}{\tau}\right)$, where $B_0 = 0.2T$ and $\tau = 50ms$
- What is the induced current in the coil?
 - Show that the total charge that flows through it is $Q = \frac{NAB_0}{R}$. Check the unit
4. (i) A solenoid of length $15cm$ has 120 turns of radius $2cm$. Ignoring end effects, what is its self-inductance?
- (ii) At what rate should change the current through it change to produce a self-induced emf of $4mV$?
5. A solenoid has 500 turns and its self-inductance is $1.2mH$
- What is the flux through each turn when the current is $2A$
 - What is the induced emf when the current changes at a rate of $35 \frac{A}{s}$
6. When the current in an inductor changes at $128 \frac{A}{s}$, the self-induced emf is $12V$. What is the self-inductance?
7. Two coils have a mutual inductance of $40mH$. What is the magnitude of the emf induced in coil 2 when the current in coil 1 changes at $25 \frac{A}{s}$
8. A solenoid with 300 turns has a length of $20cm$ and a radius of $1.8cm$. For what current will the energy density within the solenoid be $8m \frac{J}{m^3}$

9. A step-down transformer has $600V$ across the primary and $120V$. The secondary has 80 turns
- (i) What is the number of turns in the primary?
10. If a load resistance $R_L = 10\Omega$ is in the secondary, what is the primary current?